Modul 3

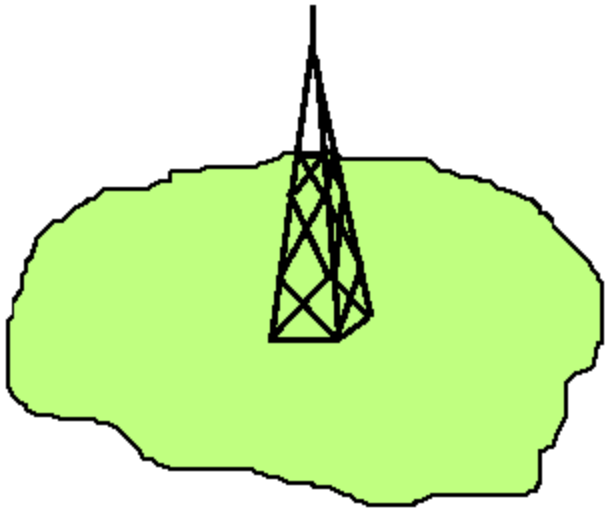
Dasar Sistem Komunikasi Bergerak Seluler

Modul 3 *Sistem Komunikasi Bergerak Seluler*

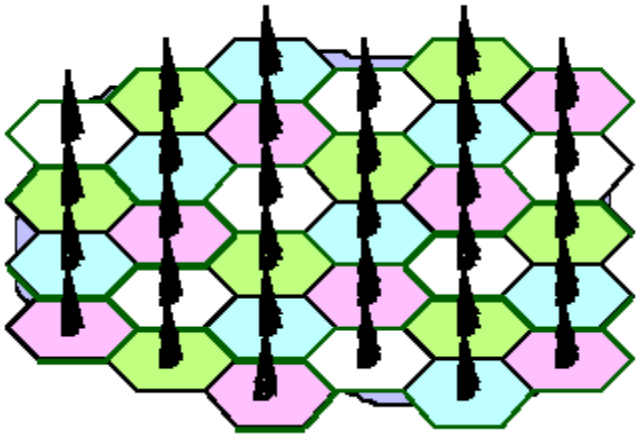
A. Pendahuluan
B. Frequency Reuse
C. Handoff
D. Channel Assignment Strategies

Yang mendasari perkembangan

- ✓ Keterbatasan spektrum frekuensi
- ✓ Efisiensi penggunaan spektrum frekuensi



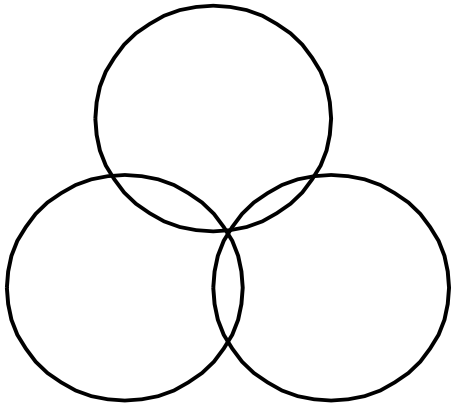
- **High power transmitter**
- **Large coverage area**



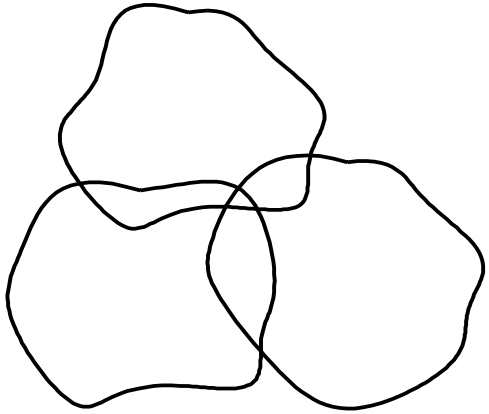
- **Low power transmitter**
- **Small coverage area**
- **Frequency reuse**
- **Handoff**
- **Central control**
- **Cell splitting to increase call capacity**

Representasi cakupan sel

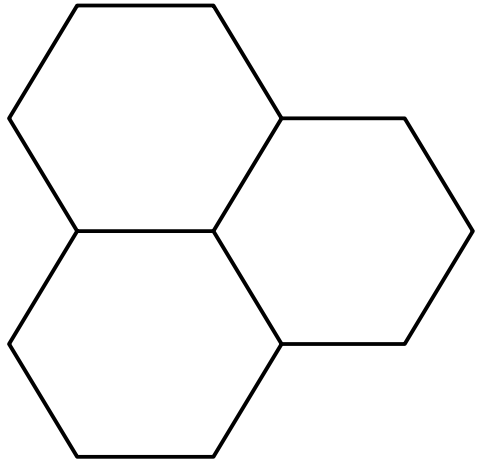
- ✓ Sel menunjukkan cakupan sinyal
- ✓ Sel berbentuk heksagonal (atau bentuk yang lain) hanya digunakan untuk mempermudah penggambaran pada layout perencanaan



SEL IDEAL

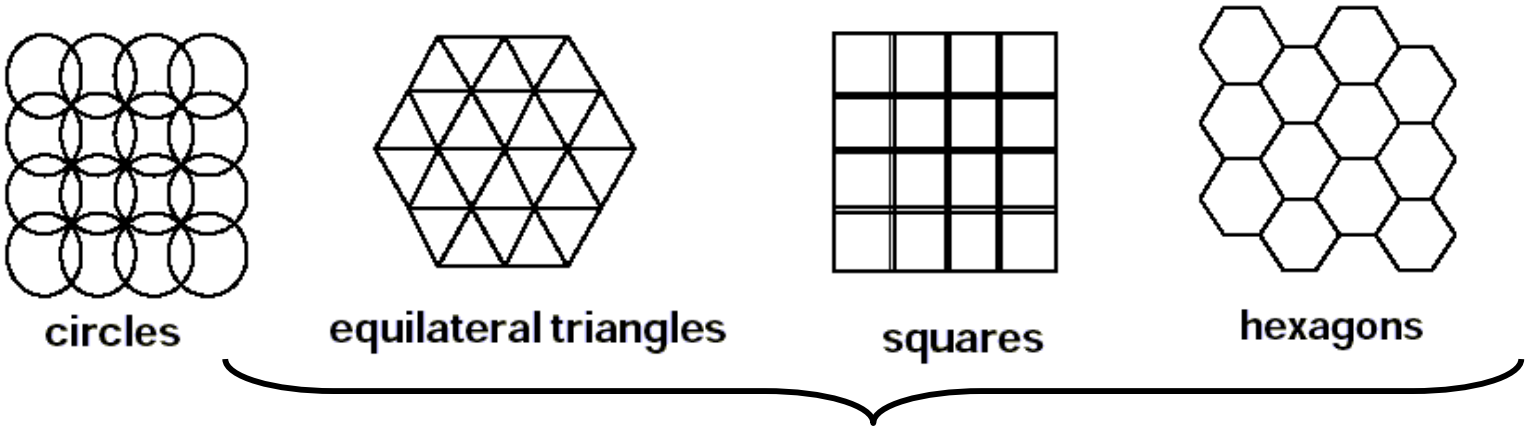


SEL REAL

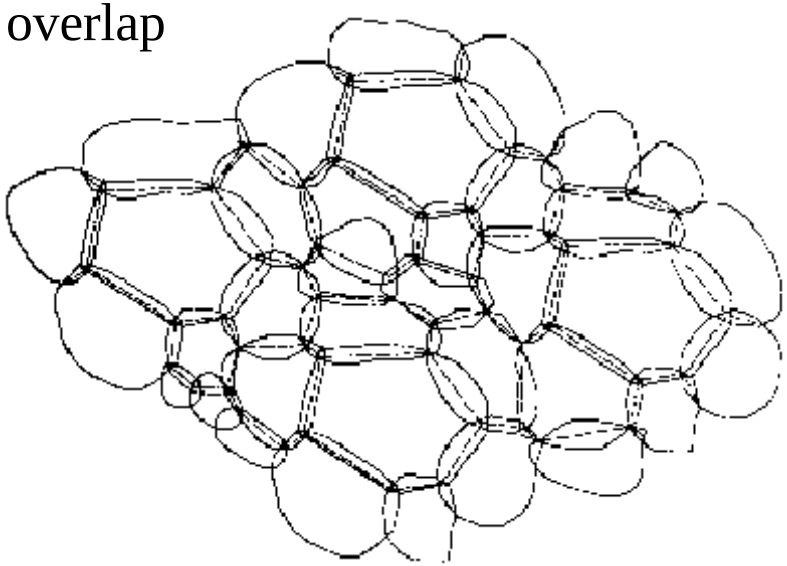


SEL MODEL

Representasi coverage sistem selular



Bentuk geometris yang meliputi keseluruhan daerah service tanpa overlap dengan luas daerah yang sama

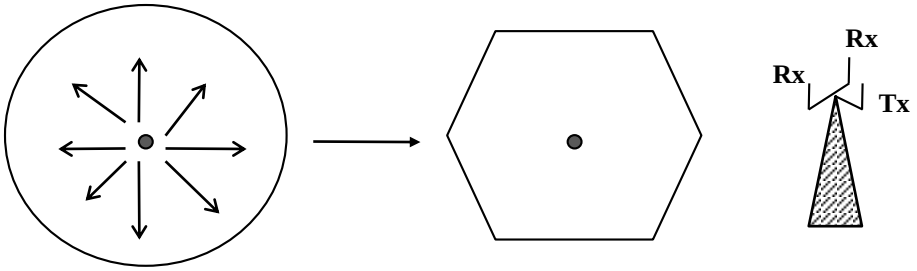


Realitas ?

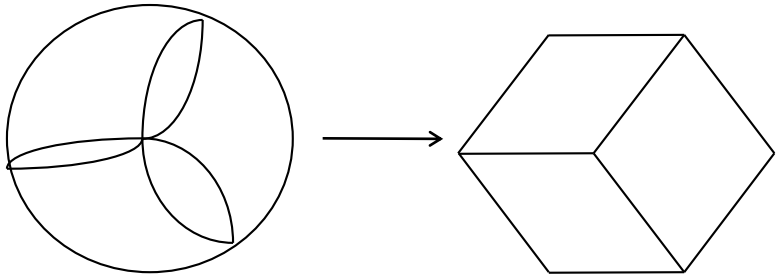
Jauh berbeda ! Grid sel teoritik digunakan untuk mempermudah penggambaran / perencanaan

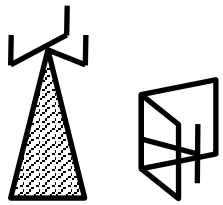
Macam-Macam Konfigurasi Sel ...

1) Omnidirectional

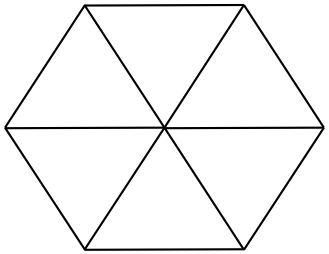


2) Sectoring 120º



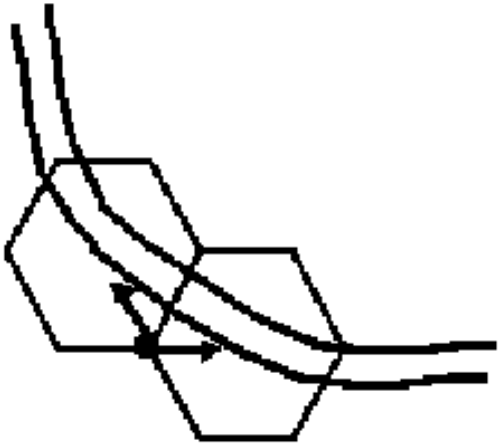


3) Sectoring 60º

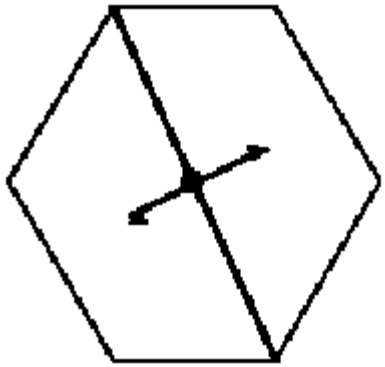


- Pada kondisi awal biasanya digunakan pola omnidirectional (tergantung demand).
- Kegunaan dari pola Sectoring
  a. Menambah kapasitas
  b. Mengurangi interferensi

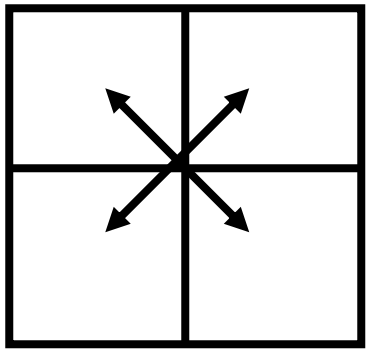
Macam-Macam Konfigurasi Sel



2-sector

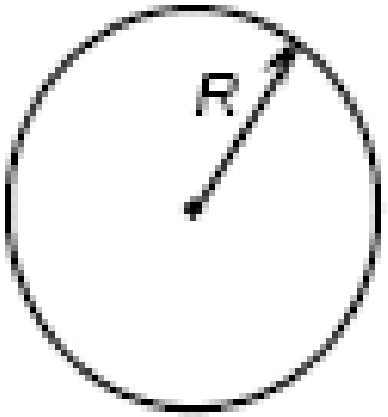


2-sector sectorized cell



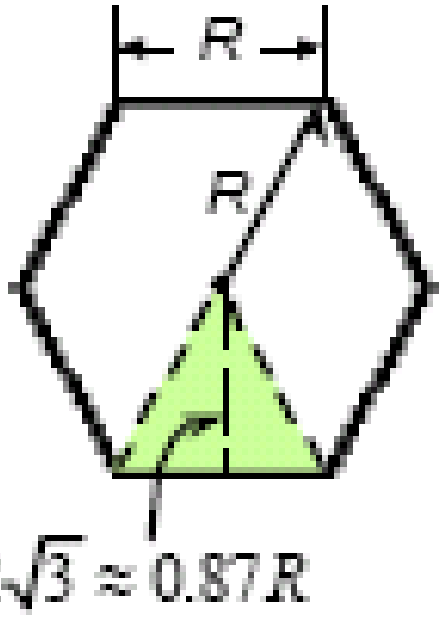
4 sector (quad sector)

Geometri Sel



$$Area_{cell} = \pi R^2 \approx 314\,R^2$$

$$Perimeter_{cell} = 2\pi R \approx 628\,R$$



$$Area_{cell} = 6 \times \tfrac{1}{2} R\sqrt{3} \times \tfrac{1}{2} R = \tfrac{3}{2} R^2 \sqrt{3} \approx 2.6R^2$$

$$Perimeter_{cell} = 6 \times R$$

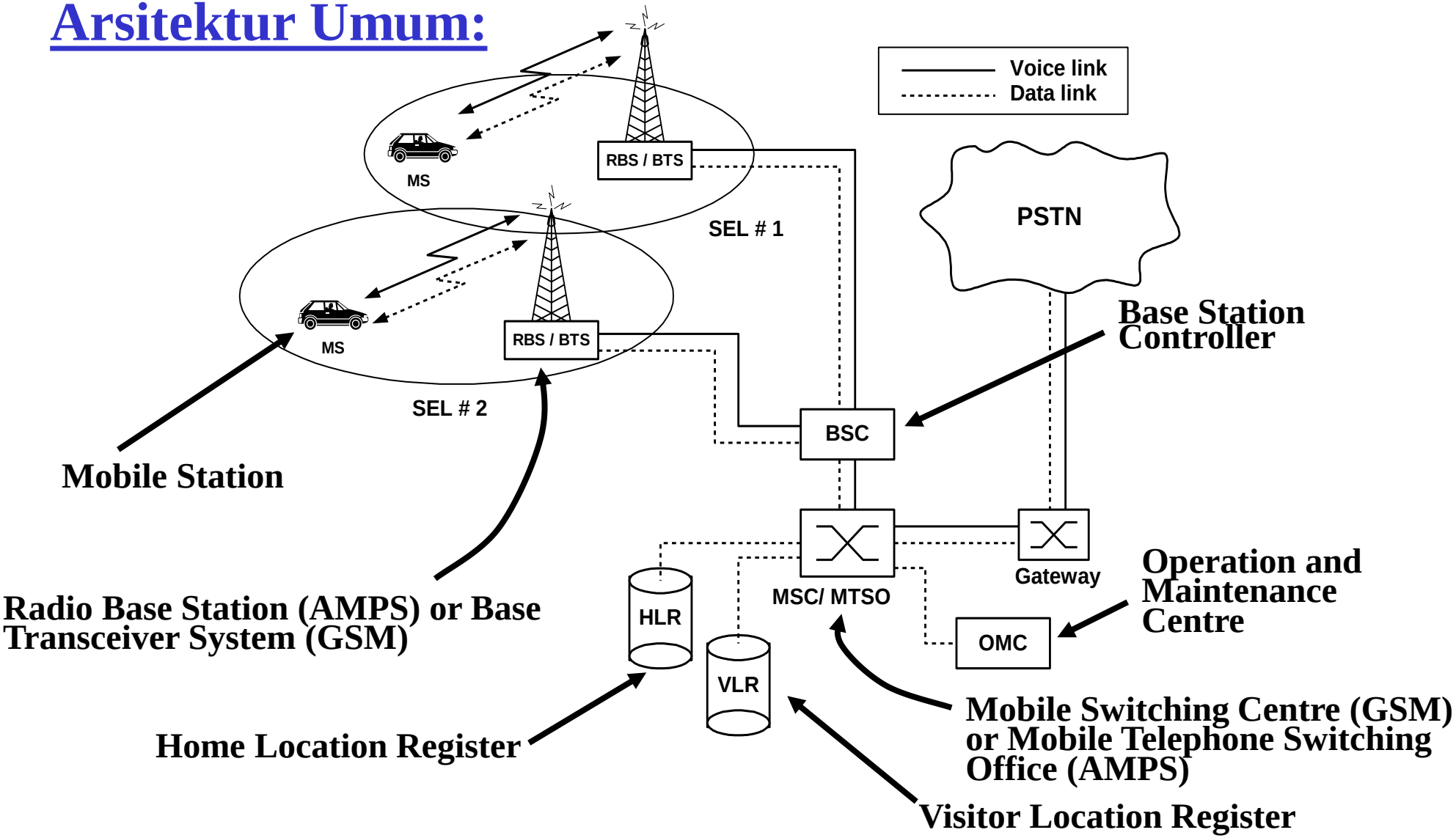
Parameter Dasar Pada Siskomber Selular

Konsep fundamental dalam teknologi komunikasi bergerak seluler:

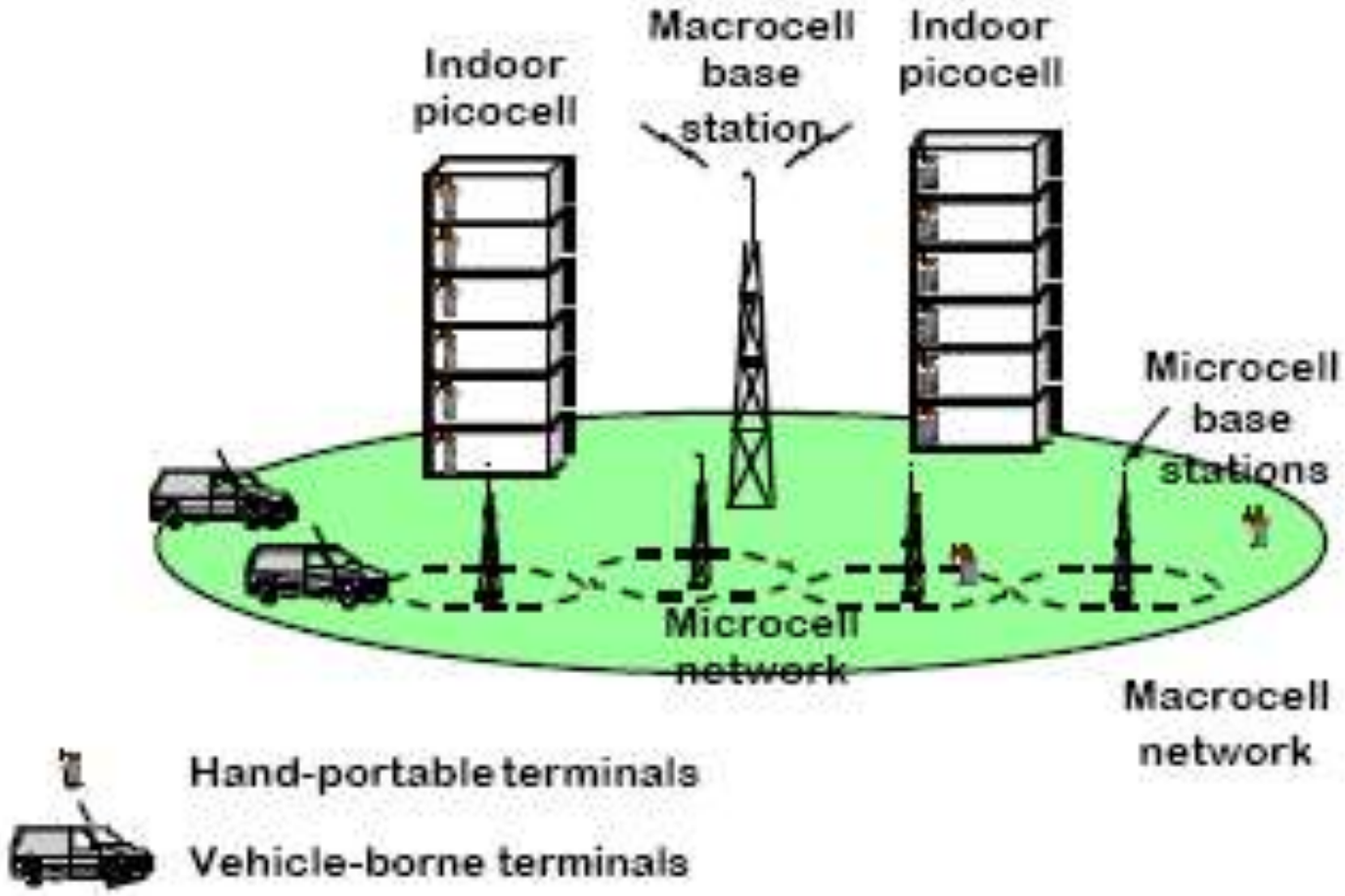
1. **Frequency Reuse**
2. **Konsep Hand Off**

- **Konsep *frequency reuse*** memungkinkan penggunaan frekuensi yang sama pada sel yang berbeda , diluar jangkauan interferensinya. Parameter yang menjadi ukuran adalah perbandingan daya sinyal / carrier terhadap total daya interferensinya
- **Sedangkan handoff** memungkinkan seorang pengguna pindah dari suatu sel ke sel yang lain tanpa adanya pemutusan hubungan. Terjadi pemindahan frekuensi / kanal secara otomatis yang dilakukan oleh sistem

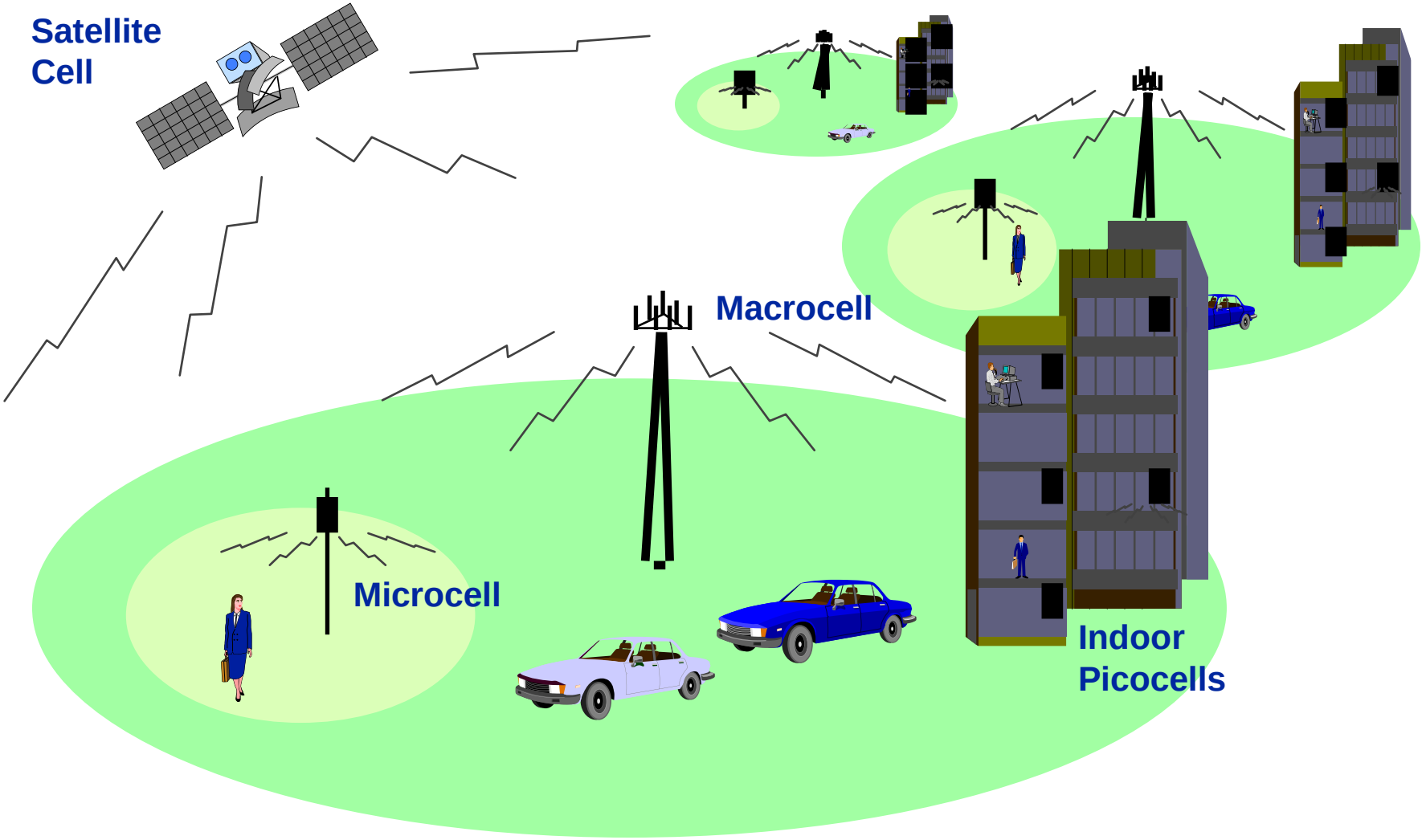
Arsitektur Umum:



Mixed Cell Architecture



Macrocell, Microcell, dan Picocell ...



MS

= Mobile Station / Mobile Unit

= Perangkat yang terdiri dari :

- Subscriber Transceiver
- Control Unit
- Antena

MTSO / MSC

= Mobile Telephone Switching Office / Mobile Switching Center.

= Merupakan pusat koordinasi dari semua cell site yang ada + berfungsi sebagai perangkat penyambung utama.

= Elemen-elemen :

- Switching Unit
- Prosesor :
  - ♣ Database processor
  - ♣ Switch processor
  - ♣ Coordination processor
- Data base unit berisi :
  - **VLR (*Visitor Location Register*)**, penyimpan data-data temporer yang masuk dari MSC lain , dan sifatnya resident
  - **HLR (*Home Location Register*)**, penyimpan data-data tetap dari pelanggan dalam MSC itu sendiri.

RBS / BTS

= Radio Base Station / Base Transceiver Station

= Merupakan perangkat transceiver yang berhubungan dari / ke pelanggan (Interface / repeater antara MS dan MSC) .

= Elemen-elemen RBS :

- Transceiver
- Control Unit / BSC / Base Station Controller
- Antena
- Data terminal

How does site location look like ?





BSC
(Switches)

site 3 sektor dengan 7 array antena tiap sektornya

jalur transmisi gelombang mikro menuju BSC



single antenna

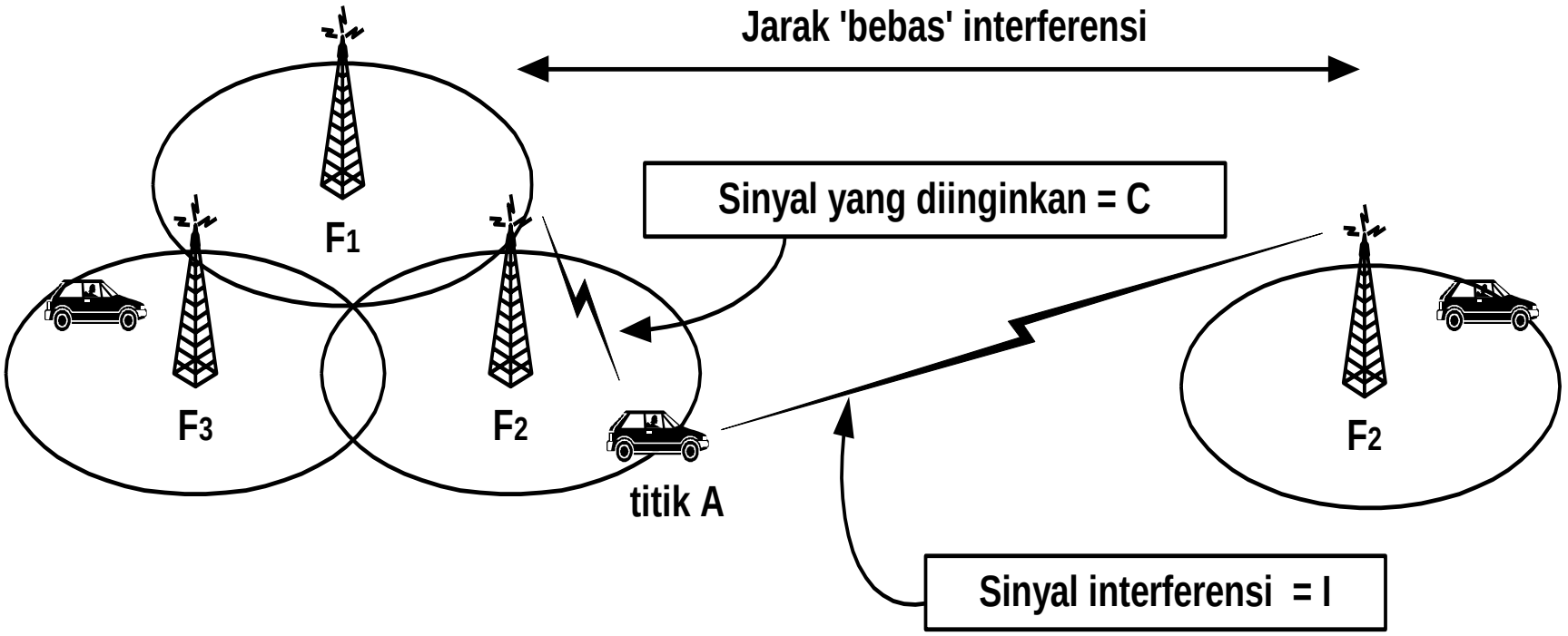
base station housing



B. Frequency Reuse

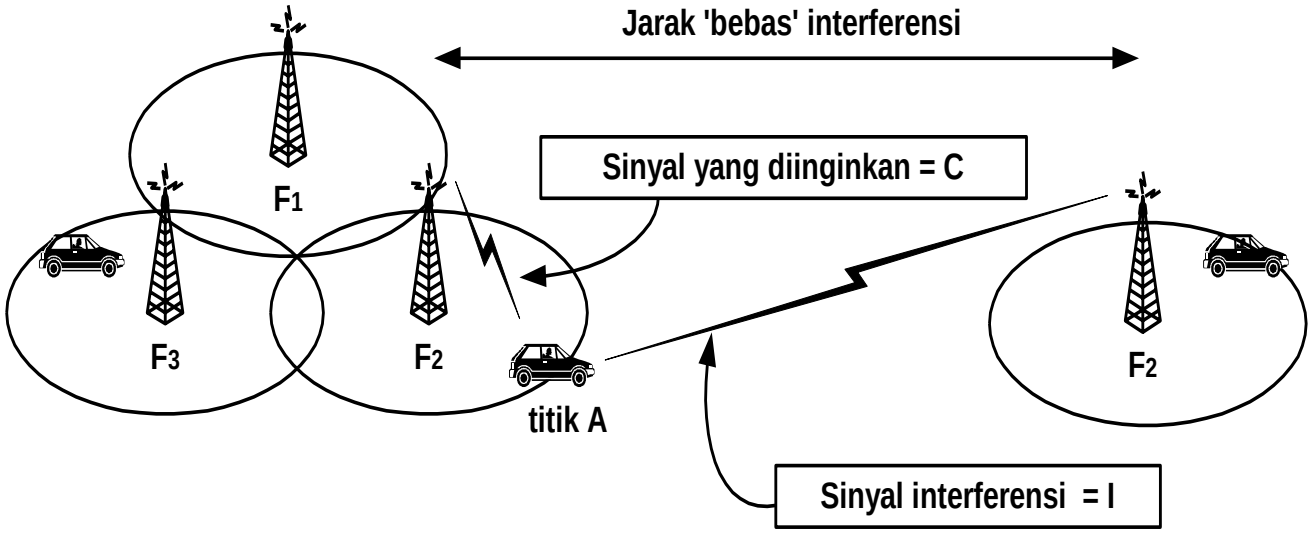
Definisi *Frequency Reuse*

*Pengulangan atau **menggunakan kembali** frekuensi yang sama pada area yang berbeda **di luar** jangkauan interferensinya*



Parameter Kinerja

→ C/I (Carrier to Interference Ratio)



- Dari gambar di atas, kondisi kasus terburuk ada pada titik A
- Pada kondisi kasus terburuk tersebut, perbandingan antara daya carrier terhadap daya interferensi (*C/I = Carrier to Interference*) **harus tetap lebih besar atau sama** dari C/I minimum yang dipersyaratkan oleh sistem seluler yang bersangkutan

→ **C/I minimum tergantung dari sistem seluler yang diimplementasikan…**

$$\frac{C}{I} = \frac{1}{N}\left[\frac{D}{R}\right]^4$$

$$\frac{D}{R} = \sqrt{3K}$$

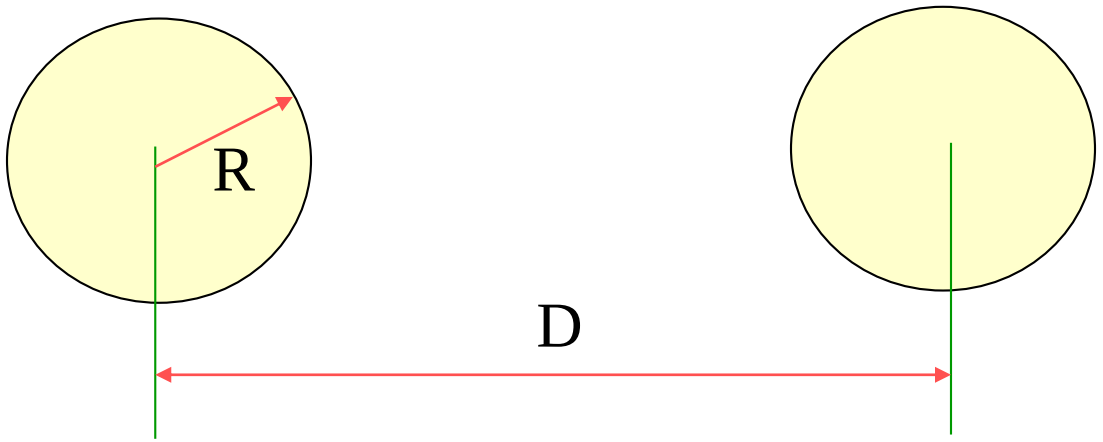
$$\frac{C}{I} = \frac{9K^2}{N}$$

AMPS, C/I = 18 dB

$$K = \sqrt{\frac{63N}{9}} = \sqrt{\frac{63.6}{9}} = 6{,}48 = 7$$

GSM, C/I = 12 dB

$$K = \sqrt{\frac{16N}{9}} = \sqrt{\frac{16.6}{9}} = 3{,}26 \approx 4$$



N = Jumlah sel penginterferensi

K = Ukuran Kluster

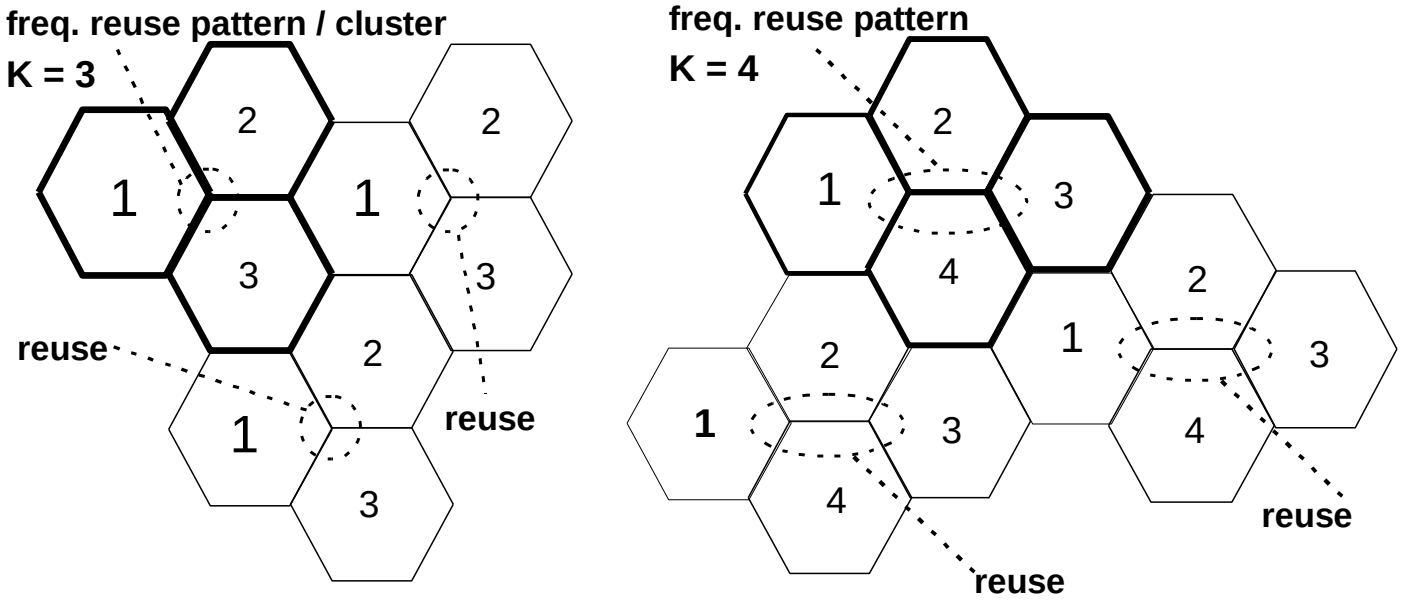
Konsep Kluster

- **Kluster** adalah sekelompok sel yang masing-masing selnya memiliki 1 set frekuensi yang berbeda dengan sel yang lain .
- **Ukuran kluster (dilambangkan = K, sering juga dilambangkan = N)** adalah jumlah sel yang terdapat dalam 1 kluster

Contoh :

K = 3 artinya terdapat 3 sel dalam 1 kluster

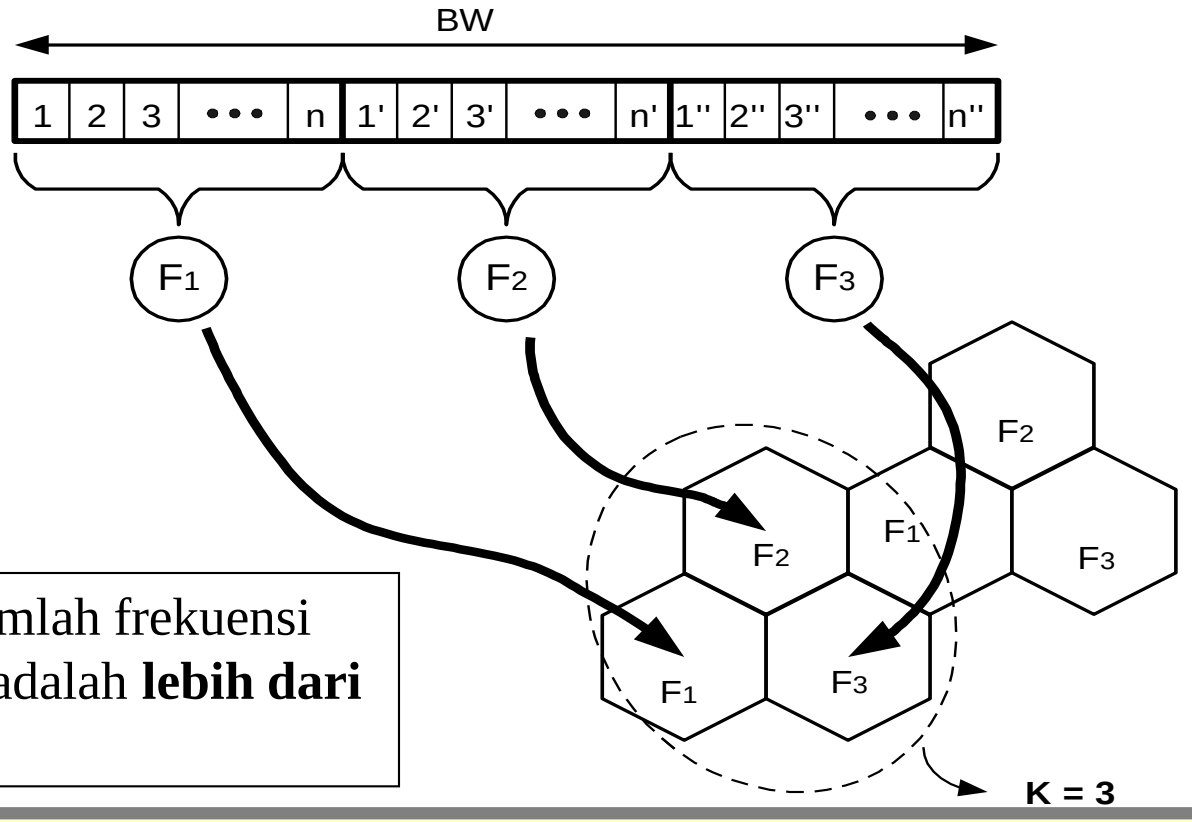
K = 4 artinya terdapat 4 sel dalam 1 kluster



Kapasitas Kanal Tiap Sel

- **Jumlah kanal tiap sel dinyatakan oleh rumus berikut :**

$$N = \frac{BW_{Alokasi}}{BW_{ch\,RF}} \frac{\text{jumlah kanal} / \text{ch RF}}{K}$$

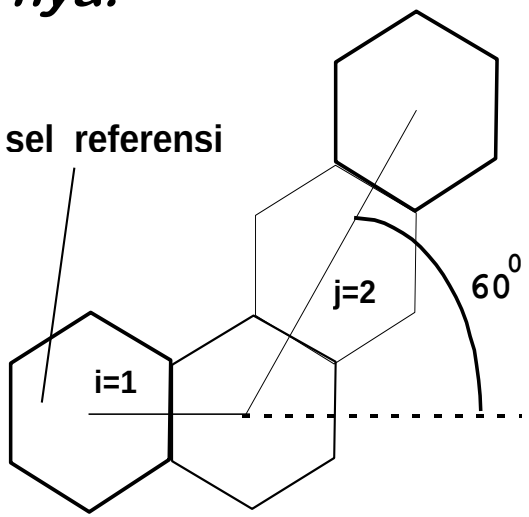


Dapat disimpulkan, jumlah frekuensi carrier dalam satu sel adalah **lebih dari satu buah…**

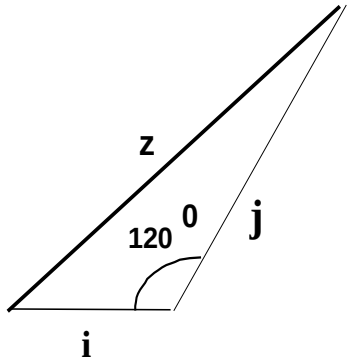
Kaidah Penentuan Nomor Sel

→ Kaidah Parameter Geser

Lalui sejauh i sel dari sel referensi sepanjang rantai heksagonalnya (garis lurus yang menghubungkan dua pusat sel), lalu berputar 60º berlawanan dengan arah jarum jam, kemudian lalui sepanjang j sel pada arah tersebut. Pada posisi akhir → disitulah letak freq. reuse nya.



i,j = 0,1,2,3, ...



$Z^2 = i^2 + j^2 - 2ij.\cos 120^o$

$Z^2 = i^2 + j^2 - 2.i.j\ (0,5)$

$Z^2 = i^2 + j^2 + i.j$

$Z^2 \cong K$ ---- K = ukuran cluster

$K = i^2 + j^2 + i.j$

untuk,

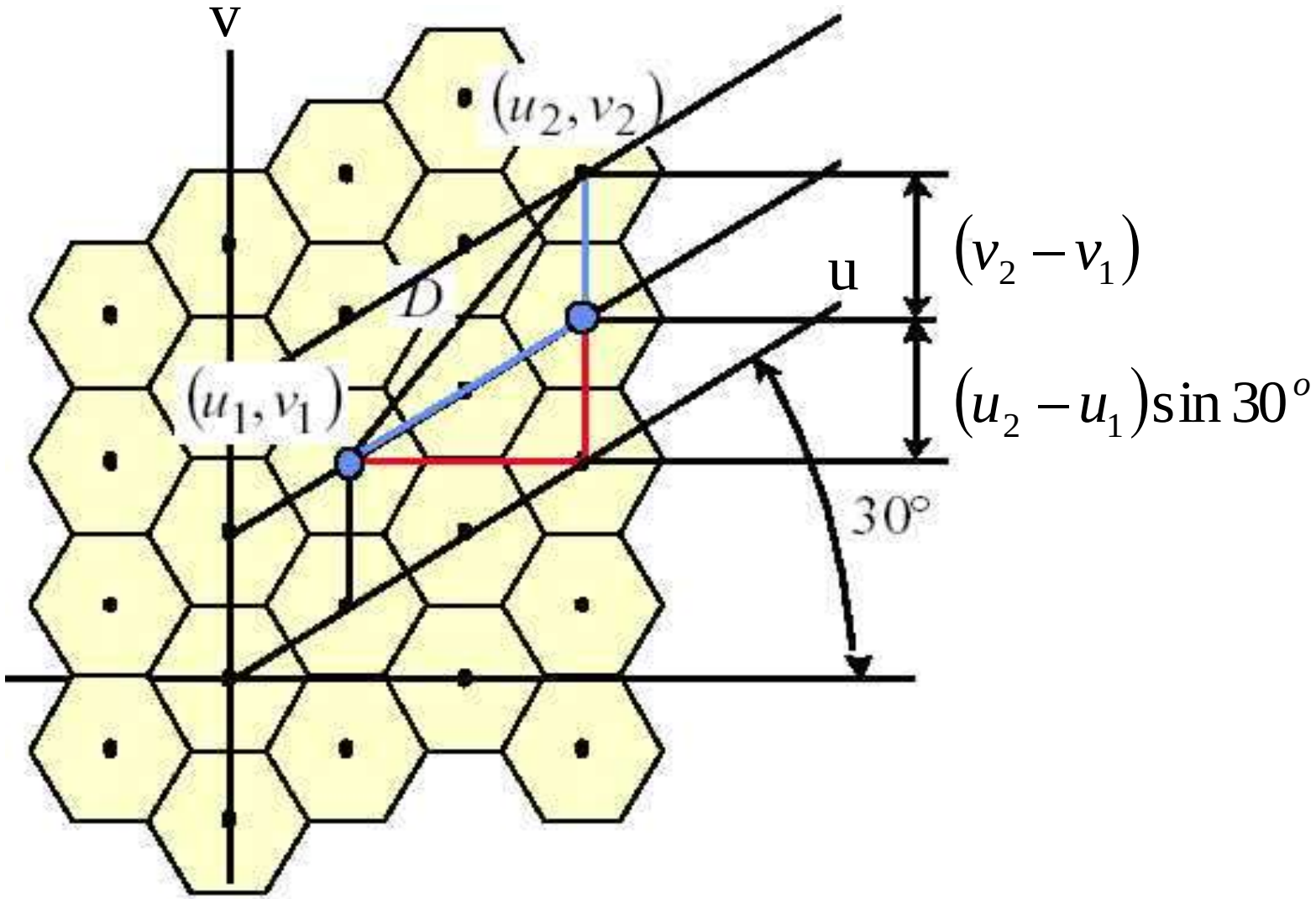
i = 1 dan j = 1 → K = 3

i = 1 dan j = 2 → K = 7

i = 0 dan j = 2 → K = 4

i = 2 dan j = 0 → K = 4

Sistem Koordinat



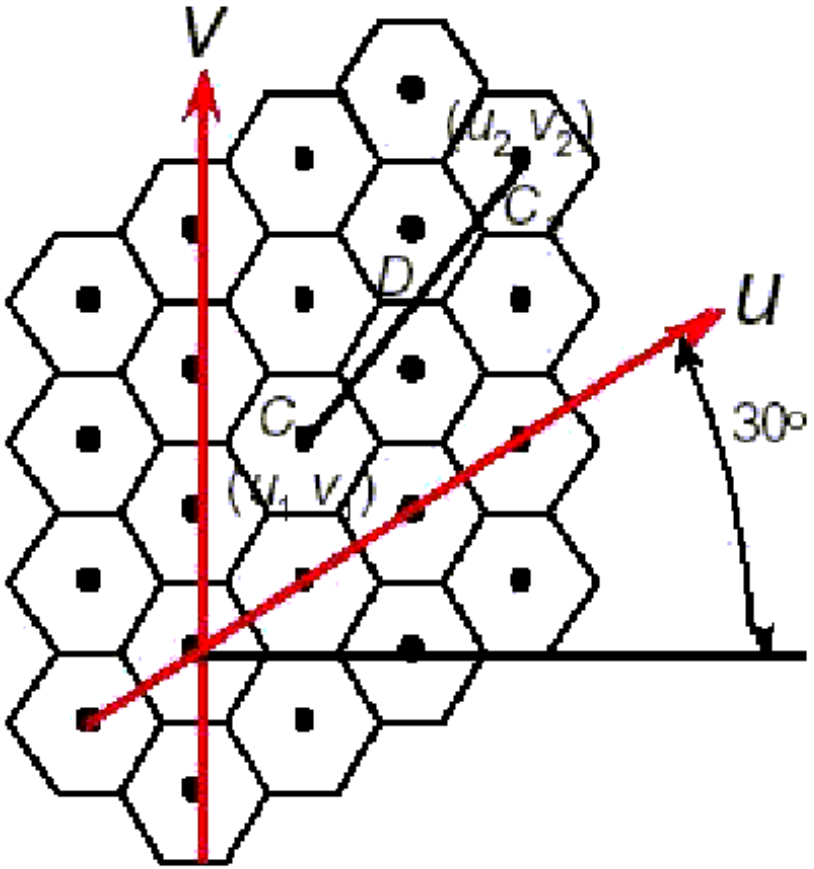
B. Frequency Reuse

Berbagai nilai kluster **K** atau **N** , yang mungkin terjadi

i	j	$N=\left(i^2+ij+j^2\right)$	$Q=D/R=\sqrt{3N}$
1	0	1	1,73
1	1	3	3,00
2	0	4	3,46
2	1	7	4,58
3	0	9	5,20
2	2	12	6,00
3	1	13	6,24
4	0	16	6,93
3	2	19	7,55
4	1	21	7,94
3	3	27	9,00

$$D = \left\{ (u_2 - u_1)^2 \left(\cos 30^o\right)^2 + \left[(v_2 - v_1) + (u_2 - u_1) \sin 30^o \right]^2 \right\}^{\frac{1}{2}}$$

$$\boxed{D = \left\{ (u_2 - u_1)^2 + (v_2 - v_1)^2 + (u_2 - u_1)(v_2 - v_1) \right\}^{\frac{1}{2}}}$$



Jika, $(u_1, v_1) = (0,0)$

$(u_2, v_2) =$ merupakan nilai integer = (i , j)

Maka,

$$\boxed{D = \sqrt{i^2 + ij + j^2}}$$

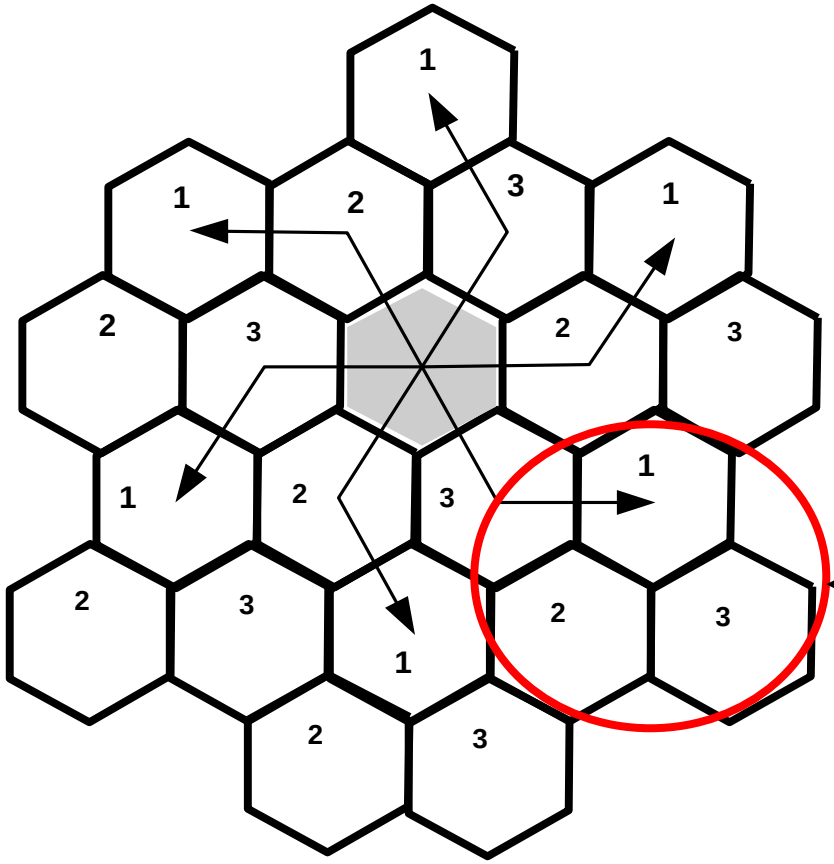
Pada contoh di samping,

i= 2 **dan** j = 1

$$D = \sqrt{i^2 + ij + j^2} = \sqrt{2^2 + 2.1 + 1^2} = 2{,}65$$

Kaidah Penentuan Nomor Sel
→ Kaidah Parameter Geser

Contoh # 1 : K = 3



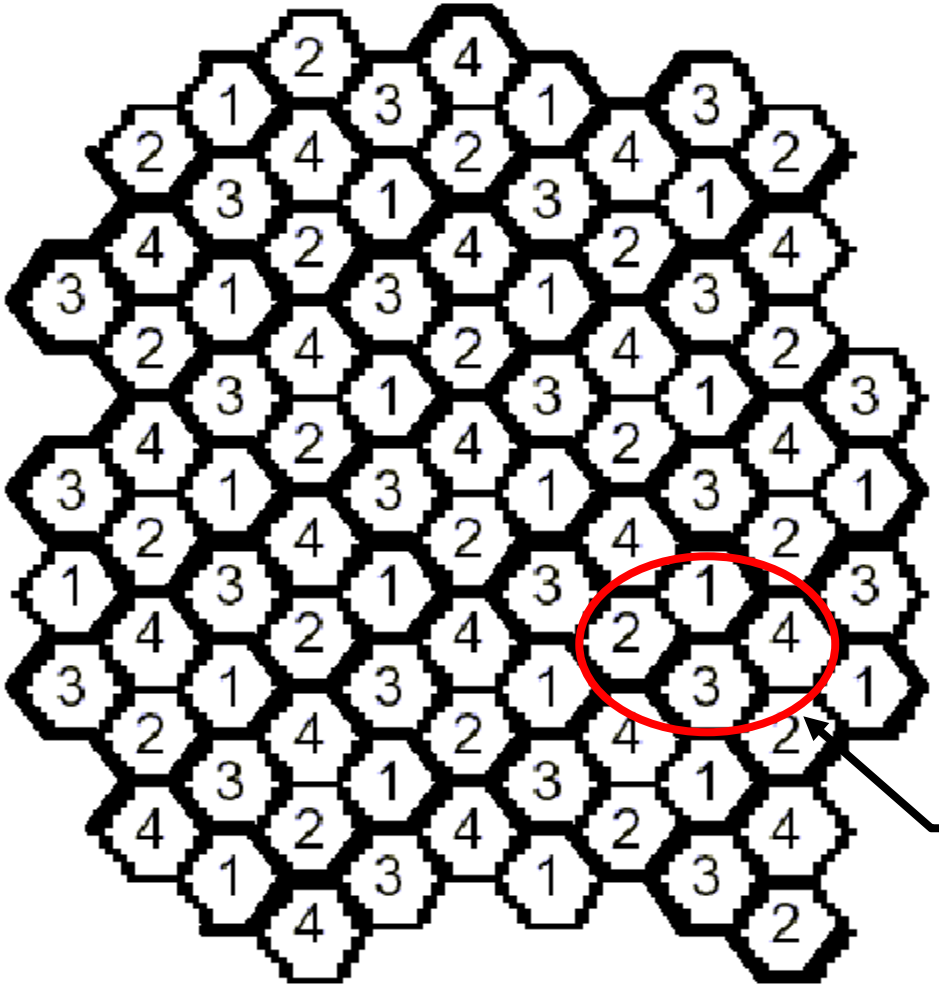
utk i = 1 dan j = 1 → K = 3

- i = 1 , j = 1
- $K = 1^2 + 1^2 + 1.1 = 3$
- Sumber interferensi maksimum = 6.

Kluster

Contoh # 2 : K = 4

Kaidah Penentuan Nomor Sel
→ Kaidah Parameter Geser



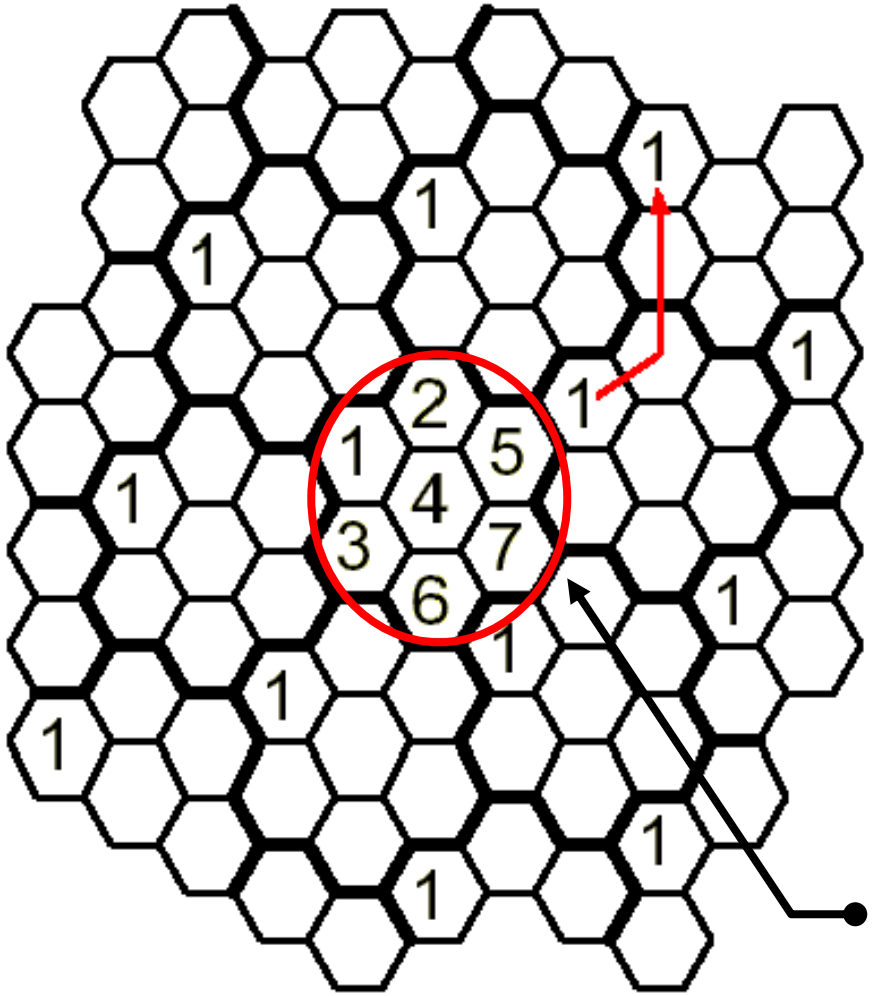
$$i = 0, j = 2$$

$$K = i^2 + ij + j^2 = 4$$

$$Q = \sqrt{3K} = 3{,}46$$

Contoh # 3 : K = 7

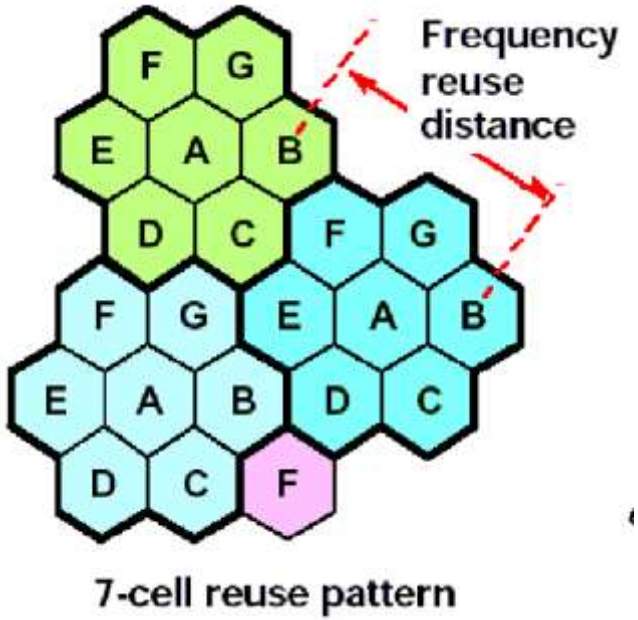
Kaidah Penentuan Nomor Sel
→ **Kaidah Parameter Geser**



$$i = 1, j = 2$$

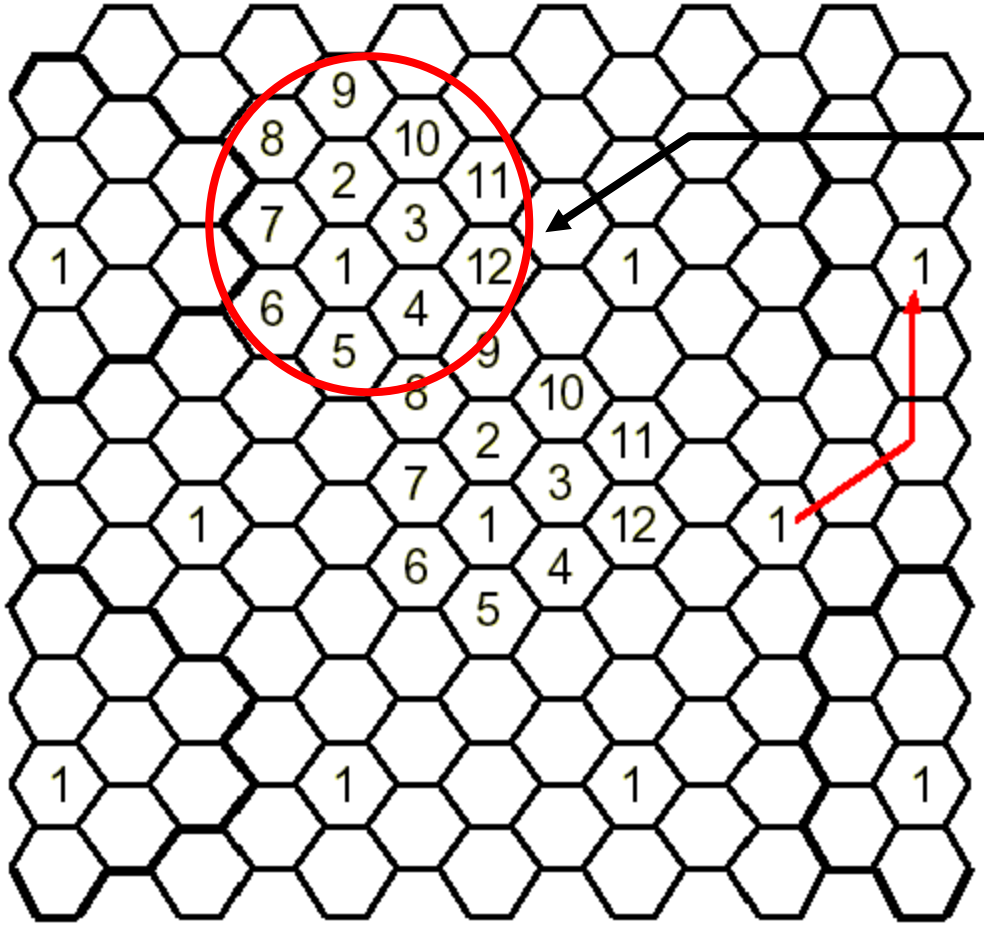
$$K = i^2 + ij + j^2 = 7$$

$$Q = \sqrt{3K} = 4{,}58$$



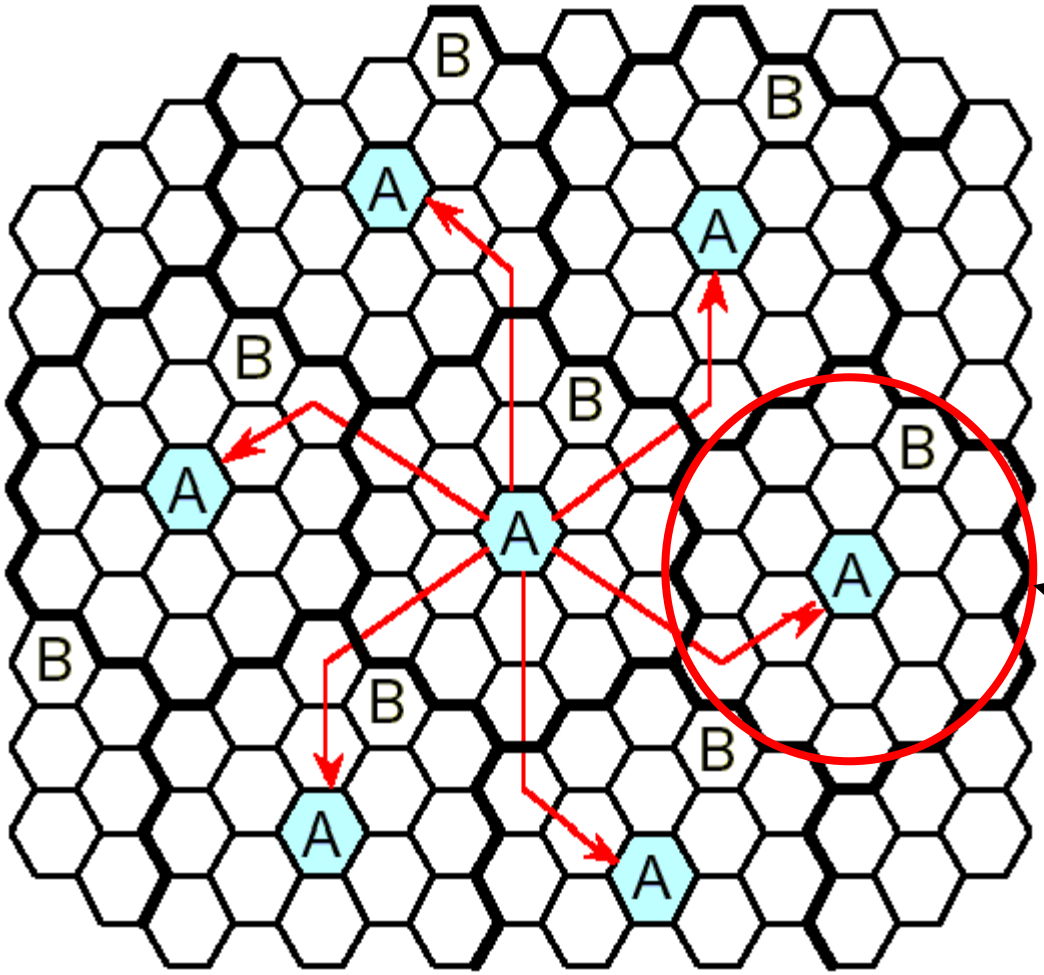
Contoh # 4 : K = 12

Kaidah Penentuan Nomor Sel
→ Kaidah Parameter Geser


9
8
10
2
11
7
3
1
12
6
4
5
9
8
10
2
11
7
3
1
12
6
4
5
1
1
1

Contoh # 5 : K = 19

Kaidah Penentuan Nomor Sel
→ Kaidah Parameter Geser

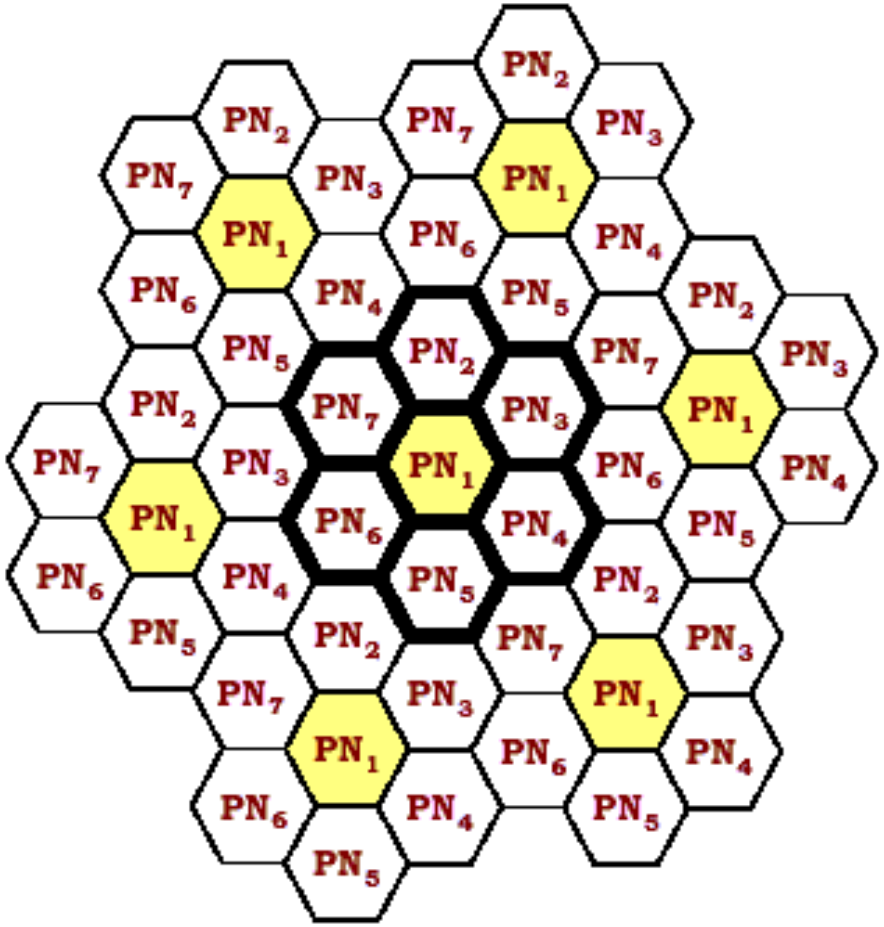


$$i = 3, j = 2$$

$$K = i^2 + ij + j^2 = 19$$

$$Q = \sqrt{3K} = 7{,}55$$

Konsepsi kluster pada CDMA…



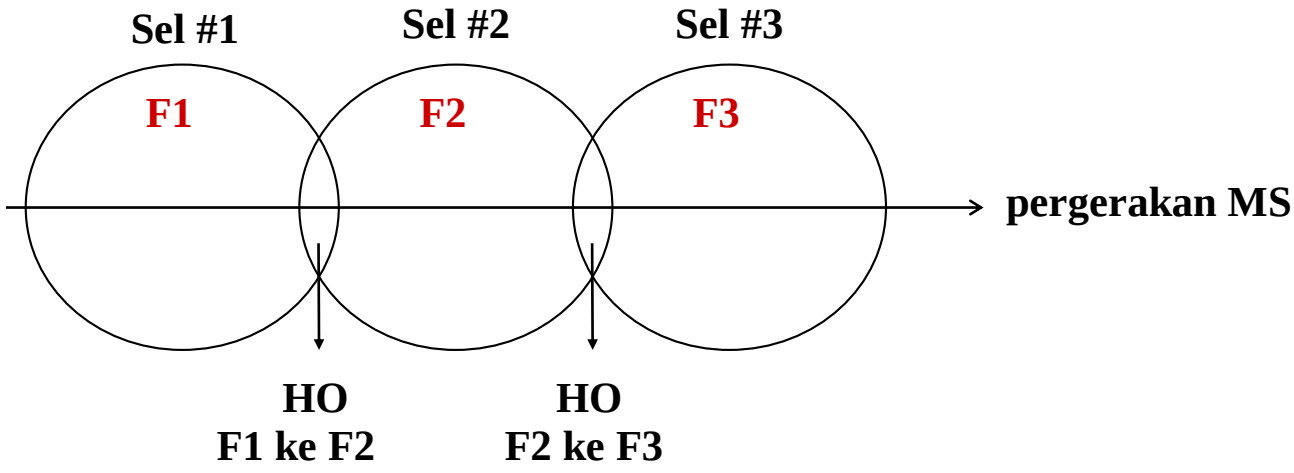
Dalam pengertian yang sama, yang sudah kita pahami…ukuran kluster di jaringan selular CDMA, $K_{CDMA} = 1$, artinya frekuensi operasi yang sama diterapkan disemua sel

Tetapi CDMA memakai konsep clustering untuk perencanaan kode PN, hal ini untuk **mencegah kemungkinan terjadinya aliasing** antar kode didalam satu sel. Pada jaringan CDMA, dikenal istilah ***PN reuse factor***

C. Handoff

Definisi

- Handoff adalah suatu peristiwa perpindahan kanal dari MS tanpa terjadinya pemutusan hubungan dan tanpa melalui campur tangan dari pemakai.
- Handoff tidak berbeda dengan handover kecuali bahwa istilah handoff digunakan di Amerika, sedangkan istilah handover digunakan di Eropa.
- Peristiwa hand over (H.O) 'umumnya' terjadi karena pergerakan MS sehingga keluar dari cakupan sel asal dan masuk cakupan sel baru.



2 Alasan dasar untuk handoff ...

- **MS keluar dari cakupan BTS (alasan klasik !)**
  - Level sinyal terima terlalu rendah
  - Bit error rate (BER) terlalu tinggi
- **Untuk keseimbangan beban jaringan**
  - Trafik disatu sel terlalu tinggi sehingga beberapa MS ‘diserahkan’ ke sel yang lain

Catatan : Standar GSM mencatat 40 alasan untuk handover !!

2 Fase handoff ...

1. **MONITORING PHASE**
   - Pengukuran kualitas sinyal dan ‘melihat’ kemungkinan radio link alternatif
   - Inisiasi handoff jika diperlukan
2. **HANDOVER HANDLING PHASE**
   - Penentuan *point of attachment* (PoA) yang baru
   - Inisiasi kemungkinan prosedur *re-routing*

Perbedaan konsepsi HO sistem seluler generasi pertama dan kedua ...

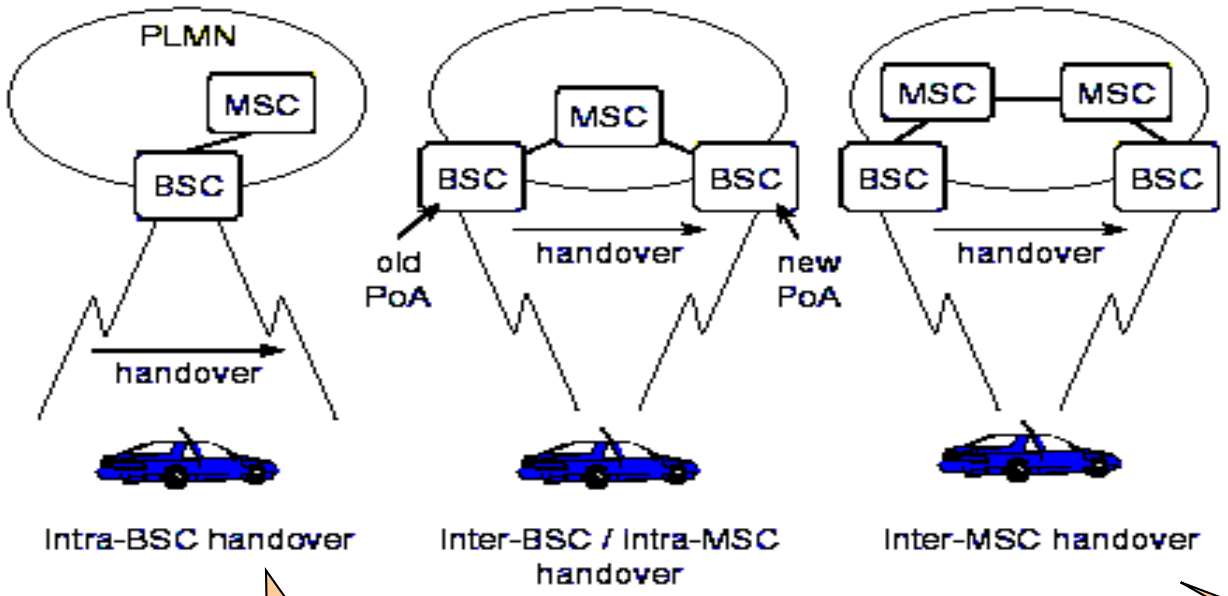
Generasi I : Sistem Analog

- Pengukuran kuat sinyal dilakukan oleh BS dan disupervisi oleh MSC
- BS secara konstan melakukan pengukuran sinyal dari tiap kanal voice
- Locator receiver mengukur kuat sinyal MS pada sel tetangga
- MSC menentukan terjadi HO atau tidak

Generasi II : Sistem Digital TDMA

- Keputusan HO dibantu MS (MAHO - *Mobile Assisted Handoff*)
- Tiap MS mengukur sinyal yang diterima dari BS yang mengelilinginya dan melaporkan ke BS-nya
- Handoff diinisiasi jika level terima dari BS tetangga mulai meningkat melebihi level sinyal dari BS-nya sendiri
- Keputusan atas dasar periode waktu atau derajat level tertentu (margin HO)
- MSC menentukan terjadi HO atau tidak

C. Handoff

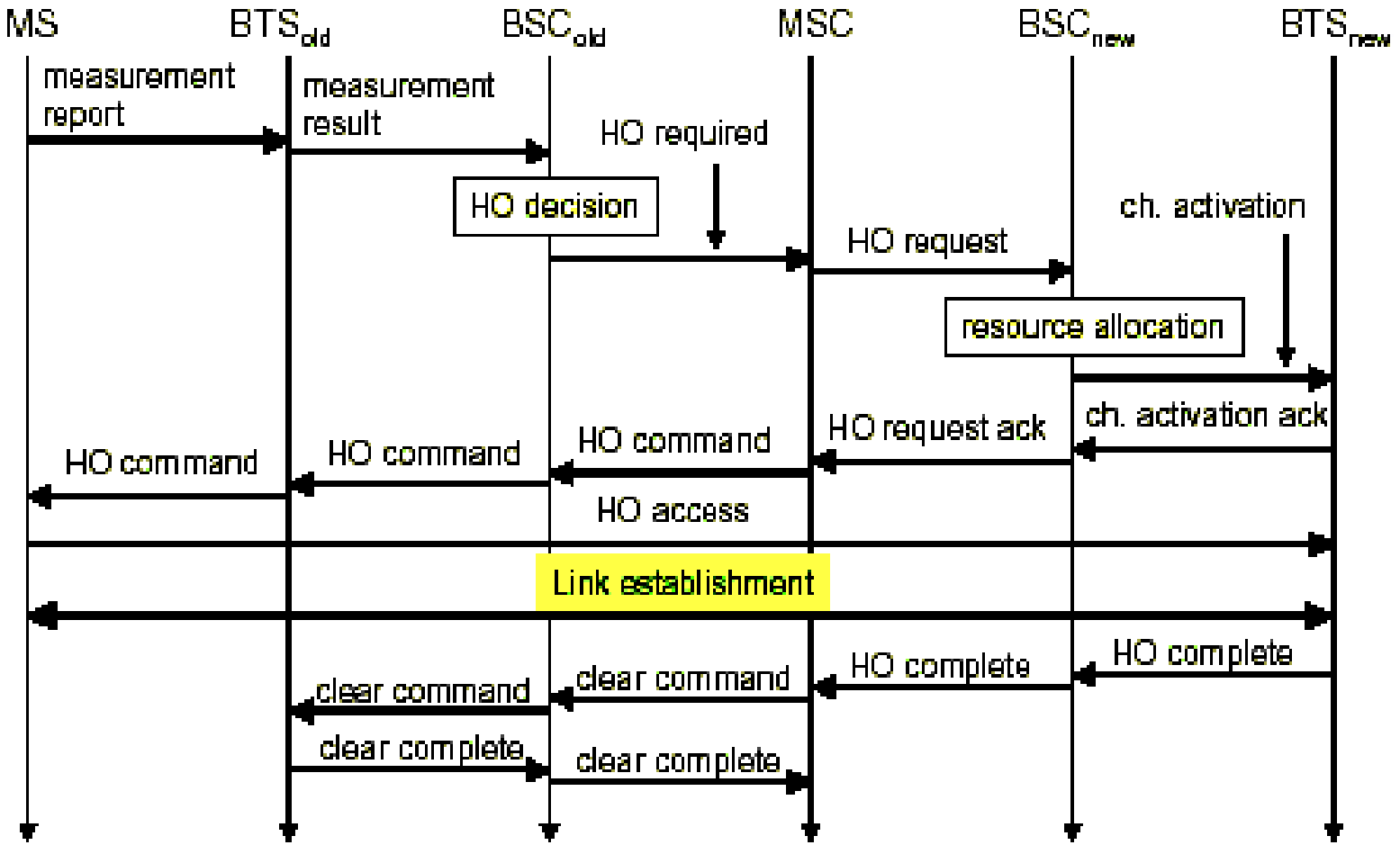


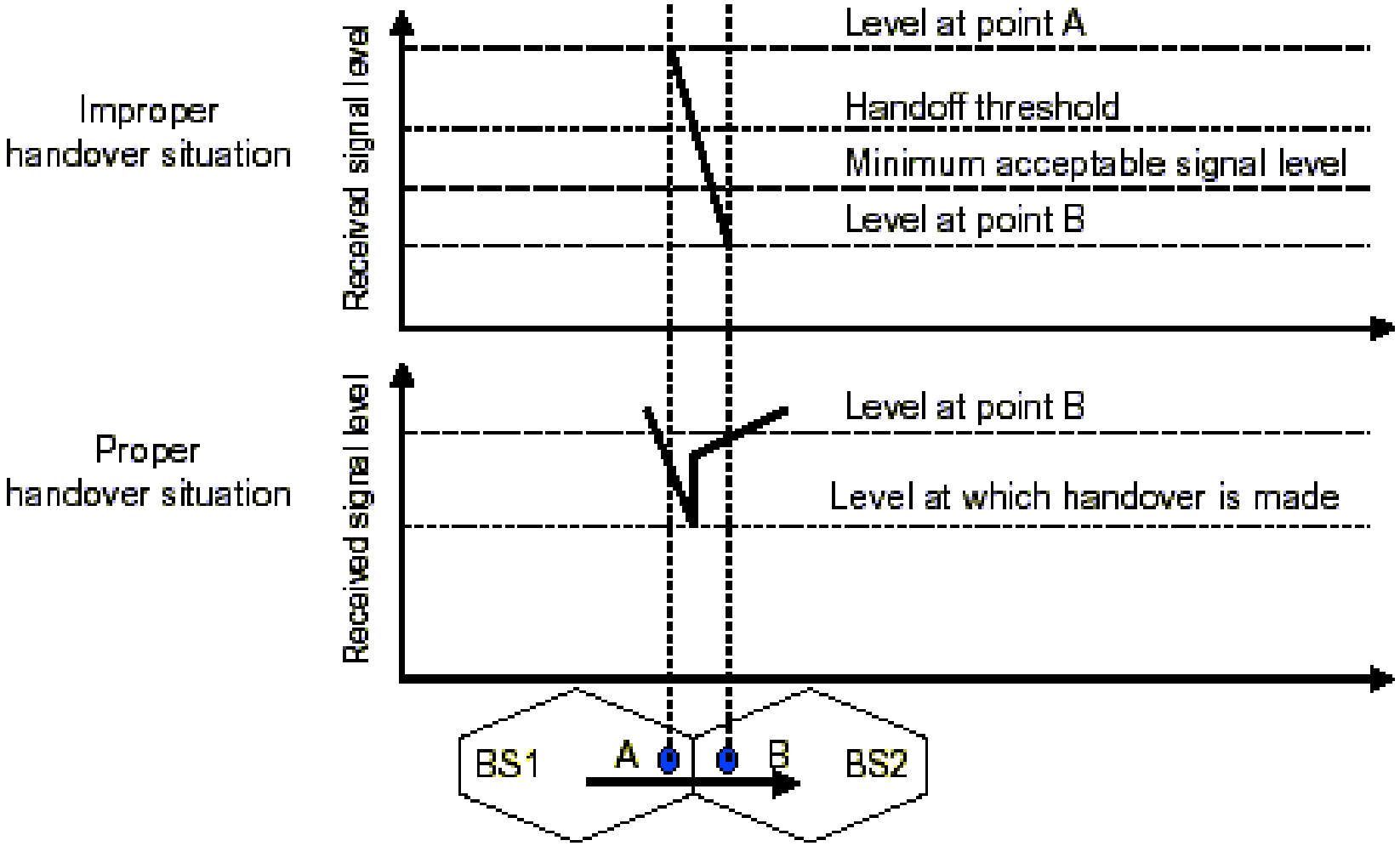
- **Sering disebabkan interferensi narrowband**

- **Kasus paling umum**
- **2 kasus : (1) inter-cell / intra-BSC, (2) inter-BSC / intra-MSC**
- **BSC melakukan operasi HO, assign kanal di sel baru dan melepas kanal lama di sel sebelumnya**

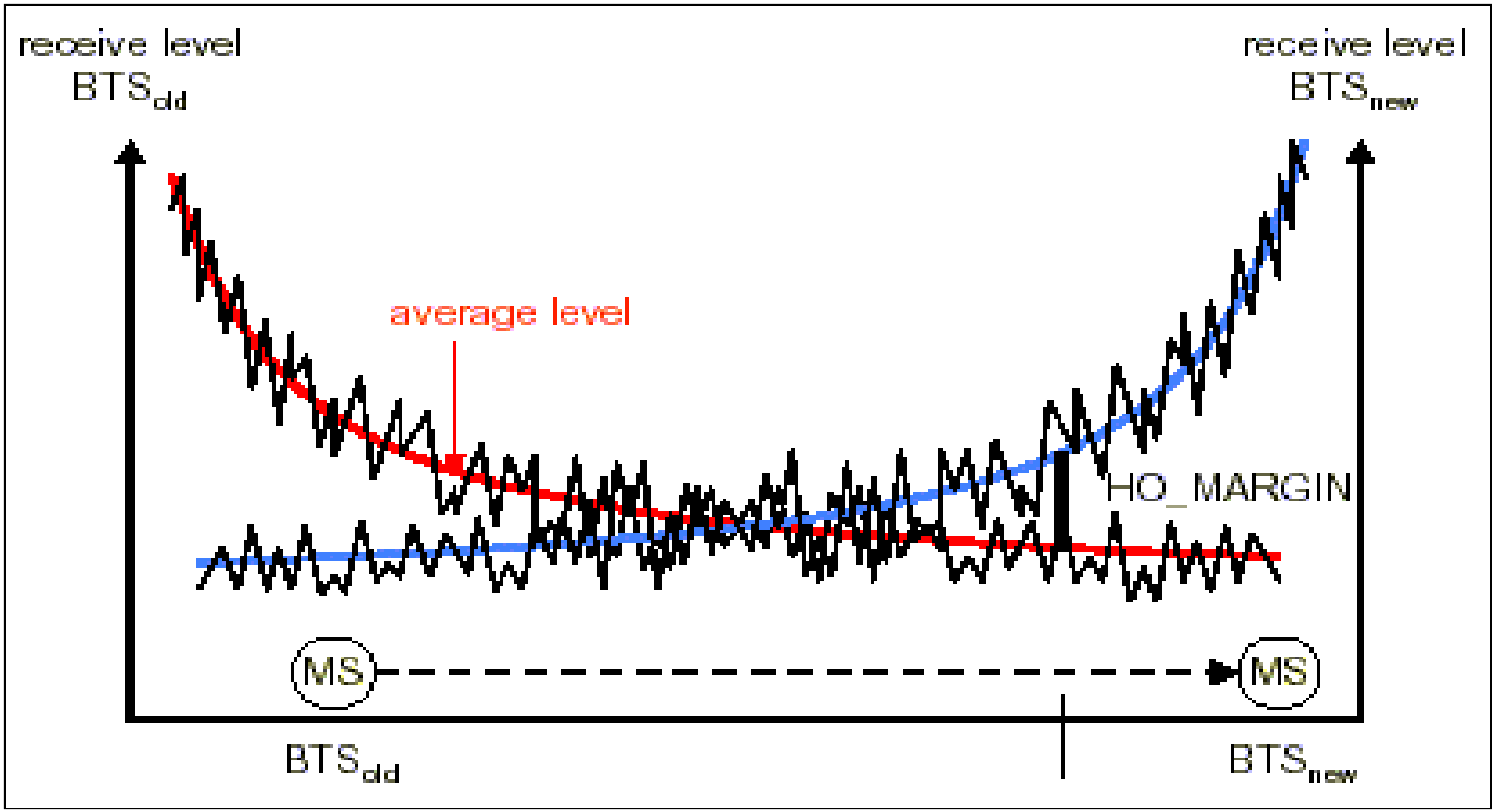
- **Dikontrol oleh kedua MSC**

Mekanisme handover ...




Improper
handover situation
Received signal level
Level at point A
Handoff threshold
Minimum acceptable signal level
Level at point B
Proper
handover situation
Received signal level
Level at point B
Level at which handover is made
BS1
A
B
BS2

C. Handoff


receive level
BTS old
receive level
BTS new
average level
HO_MARGIN
MS
MS
BTS old
BTS new

D. Channel Assignment Strategies

Channel assignment / channel alocation :

Proses pengalokasian/ pemberian kanal trafik

Kanal trafik perlu diberikan kepada user berkaitan dengan :

- Panggilan baru di dalam sel
- Kejadian handover

Ada bermacam-macam skema algoritma pengalokasian kanal (*channel allocation scheme*) , dan **skema pengalokasian kanal dapat mempengaruhi performansi sistem !**

3 kelas channel assignment

a. Fixed channel assignment
Kanal yang disediakan dalam 1 sel / sektor tertentu, modifikasi dari metoda ini adalah *borrowing scheme*

b. Dynamic channel assignment
Kanal tidak dialokasikan dalam sel/sektor secara permanen. MSC mengalokasikan kanal berdasarkan : probabilitas blocking mendatang di semua sel, jarak reuse,C/I, *cost factor* , dsb **meningkatkan kompleksitas sistem !!**

c. Others : hybrid kedua hal diatas, *scheduling*, *prediction*, *prioritisation*

Asistensi # 1

- **Bahas berbagai strategi channel assignment**

  → Lihat paper berikut :

  Katzela, Naghshineh, " ***Channel Assignment Schemes for Cellular Mobile Telecommunication Systems – A Comprehensive Survey*** "

Example For Frequency Planning

Outline

- **Introduction**
- **Multiple Reuse Pattern**
- **Frequency Plan Basics**
- **Frequency Grouping**
- **Frequency Plan Constraints**

Introduction

- **Frequency planning is probably the most challenging and time consuming part of the radio network design. The best way to handle this is to use an automatic frequency planning tool.**
- **To get high efficiency of the available frequencies it is important to analyse the network performance to pinpoint the weaknesses, e.g. congestion, and increase the capacity in areas where mostly needed.**
- **To increase capacity by using a tighter frequency reuse than a nominal 4/12 pattern, which is the most commonly used pattern today, separation of the BCCH band and the TCH band should be introduced. By using different reuse patterns on BCCH channels and TCH channels the reuse pattern can go from 4/12 to e.g. a 9 re-use.**
- **The BCCH carrier has to be planned in a careful manner and a 12 re-use pattern shall be used as a minimum, in order for the frequency plan to perform well. The TCH carriers can be planned in different reuse patterns depending of how many frequencies that are available and on the amount of hardware installed in the base stations.**

Introduction

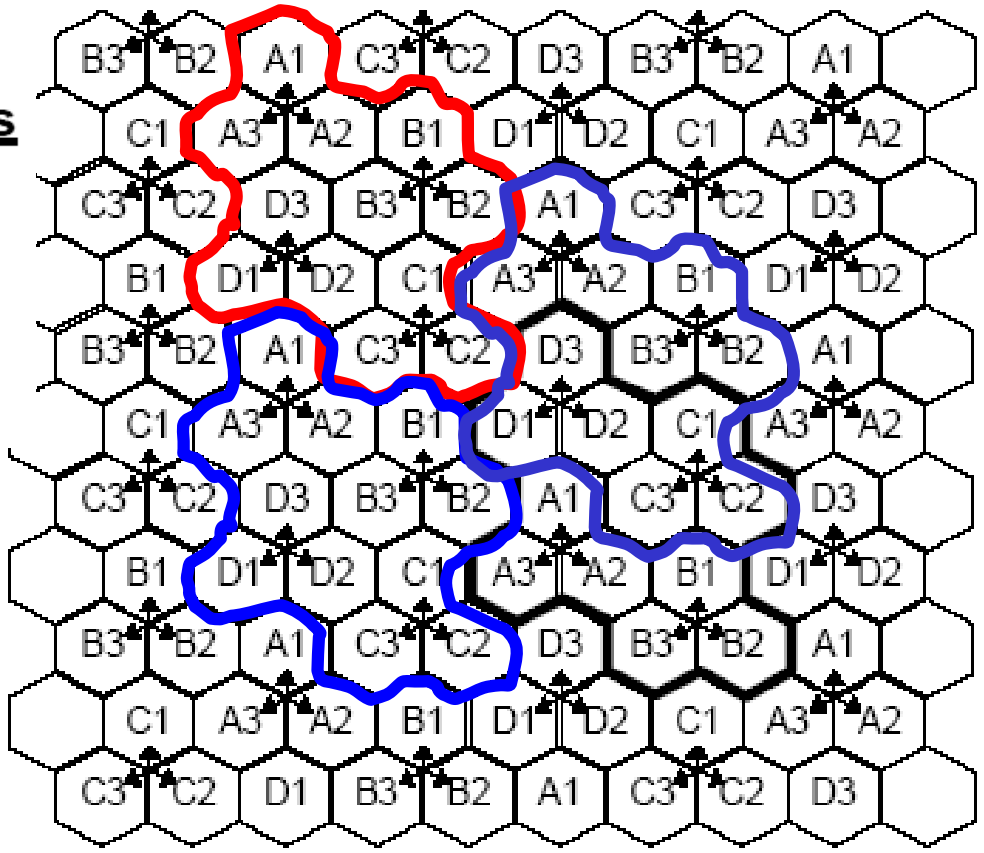
- **All features available to decrease or average the interference level in the network should be used (e.g. Frequency hopping, Dynamic Power Control, DTX). The most important feature is frequency hopping and the hopping should be made over as many frequencies as possible.**
- **The TCH frequencies are planned in different layers with independent re-use schemes. The method is called MRP (Multiple Re-use Pattern) and can be described according to the reuse patterns of the different layers. First the reuse of the BCCH, followed by the different TCH patterns (e.g. 12/8/7).**
- **An MRP of 12/8/7 has a re-use of 12 on the BCCH carriers and a re-use of 8 on the first TCH carriers and 7 on the third. This particular scheme is equivalent to a nominal 3/9 reuse pattern. If there are 32 frequencies available and four TRXs installed in three sector sites an MRP of 12/8/5/5 or 12/8/6/4 can be used which is equivalent to a nominal reuse of 8 (32/4).**

Introduction

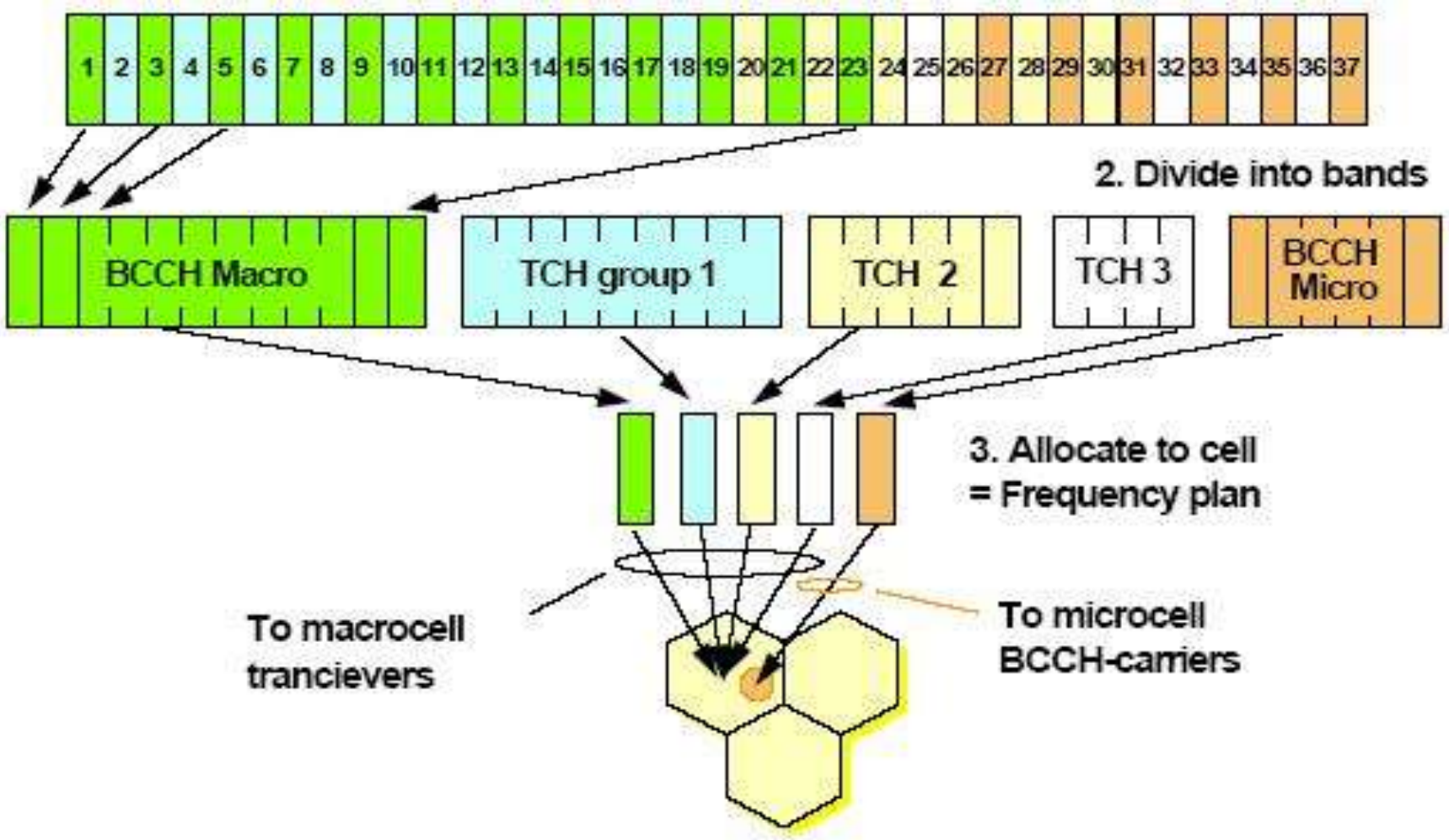
- **Before the actual frequency planning is started, the available frequencies are normally arranged into frequency groups. The most common re-use pattern today is a 4/12 clover-leaf pattern**

Frequency grouping in 4/12 re-use patterns

A1	B1	C1	D1	A2	B2	C2	D2	A3	B3	C3	D3
1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	and so on - - -				- - -	- - -	- - -	



MRP : Multiple Reuse Pattern



Example : GSM Channel Numbering

- GSM900

  $F_U(n) = 890 + 0.2 \times n$ (MHz)

  $F_D(n) = F_u(n) + 45$ (MHz) $1 \le n \le 124$

- E-GSM900

  $F_U(n) = 890 + 0.2 \times (n-1024)$ (MHz) $974 \le n \le 1023$

  $F_D(n) = F_u(n) + 45$ (MHz)

  n is called *Absolutely Radio Frequency Channel Number* (ARFCN)

- GSM1800

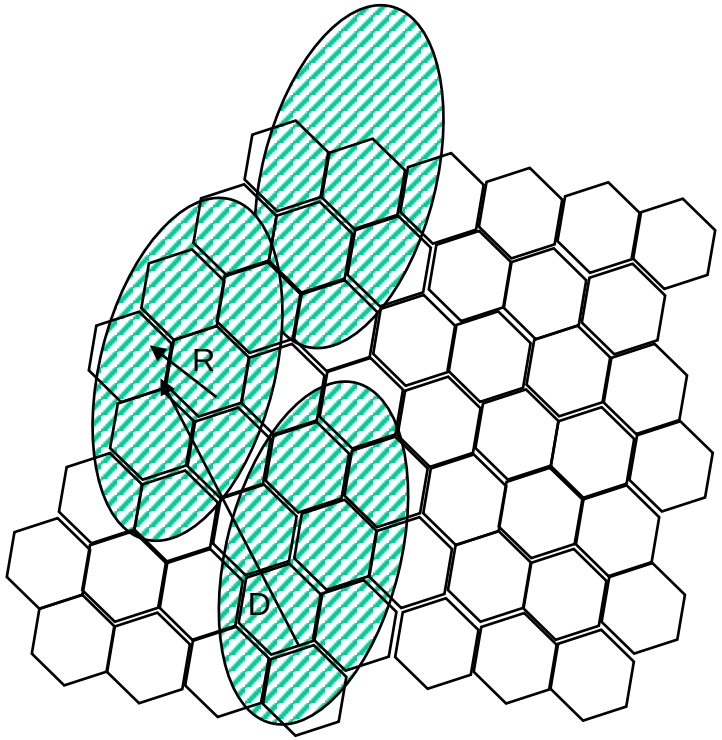
  $F_u(n) = 1710.2 + 0.2 \times (n-512)$ (MHz)

  $F_D(n) = F_U(n) + 95$ (MHz) $512 \le n \le 885$

  374 channels

Frequency Plan – Basics

- **Tighter re-use of own frequencies**
  ⇒ more capacity
  ⇒ more interference
- **Target**
  - to minimise interferences at an acceptable capacity level
- **First when a complete area has been finalised**
- **Automatic frequency planning tools**



Frequency Plan Basics

- **Why frequency re-use ?**
  - **8 MHz = 40 channels à 7 traffic timeslots = 280 users**
  - **max. 280 simultaneous calls??!**
- **Limited bandwidth available**
  - **Re-use frequencies as often as possible**
  - **Increased capacity**
  - **Increased interferences**
- **Trade-off between interference level and capacity**
- **Allocate frequency combination that creates least overall interference conditions in the network**



Interference is unavoidable
⇒ minimise total interferences in network

Frequency Planning Criteria

- The frequency planning criteria include the configuration and frequency allocation aspects. The configuration aspects consider the:
  - Frequency band splitting between the macro and micro base stations,
  - Frequency band splitting between the BCCH and TCH layers,
  - Frequency band grouping and
  - Different frequency reuse factors for different TRX layers.

- Frequency allocation aspects include frequency planning thresholds (QOS requirements)
  - C/I requirements
  - Percentage of co-channel and adjacent channel interference

Frequency Band Splitting

Macro - Micro

- needed because of inaccurate coverage predictions between macro and micro layers
- not needed if accurate coverage predictions available in the future

BCCH - TCH

- needed to ensure a good quality on BCCH frequency (in order to ensure signalling)

Frequency Band Grouping

Frequency grouping

+ Frequency hopping (coherence bandwidth)
+ Intermodulation
+ Frequencies assigned to all TRX layers at one time
+ Frequencies evenly used
- Limitations for automatic frequency planning algorithms
- Fixed frequency reuse factor

	f1	f2	f3	f4	f5	f6	f7	f8	f9	f10	f11	f12	f13	f14
BCCH	1	2	3	4	5	6	7	8	9	10	11	12	13	14
2. TRX	15	16	17	18	19	20	21	22	23	24	25	26	27	28
3. TRX	29	30	31	32	33	34	35	36	37	38	39	40	41	42

Different Frequency Reuse Factors for Different TRX Layers

Frequency planning for different TRX layers

- different freqency reuse factors for different TRX layers
- frequency planning for different layers

	f1	f2	f3	f4	f5	f6	f7	f8	f9	f10	f11	f12	f13	f14	f15
BCCH	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
2. TRX	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1. Micro	31	32	33	34	35	36	31	32	33	34	35	36	31	32	33
2. Micro	37	38	39	40	41	42	37	38	39	40	41	42	37	38	39

Frequency Allocation Threshold

C/I requirements

- C/Ic = 12 dB or 15 dB (save value), C/Ia = -6 dB (Note Overlay-Underlay concepts)

Interference probability

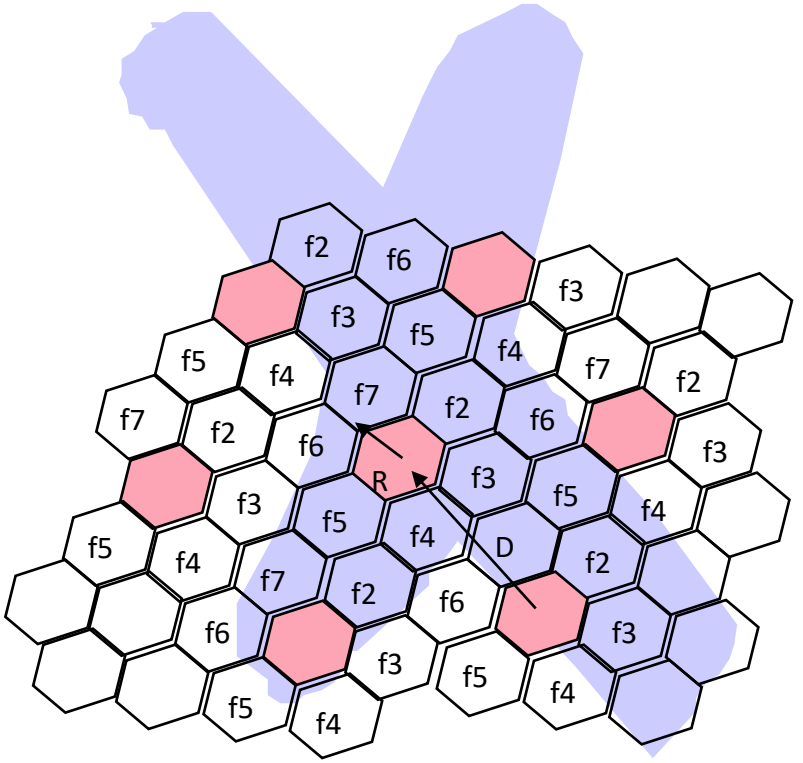
- 2% co-channel and 5% adjacent channel interference

Frequency separations

- cell/site separations
- combiner limitations

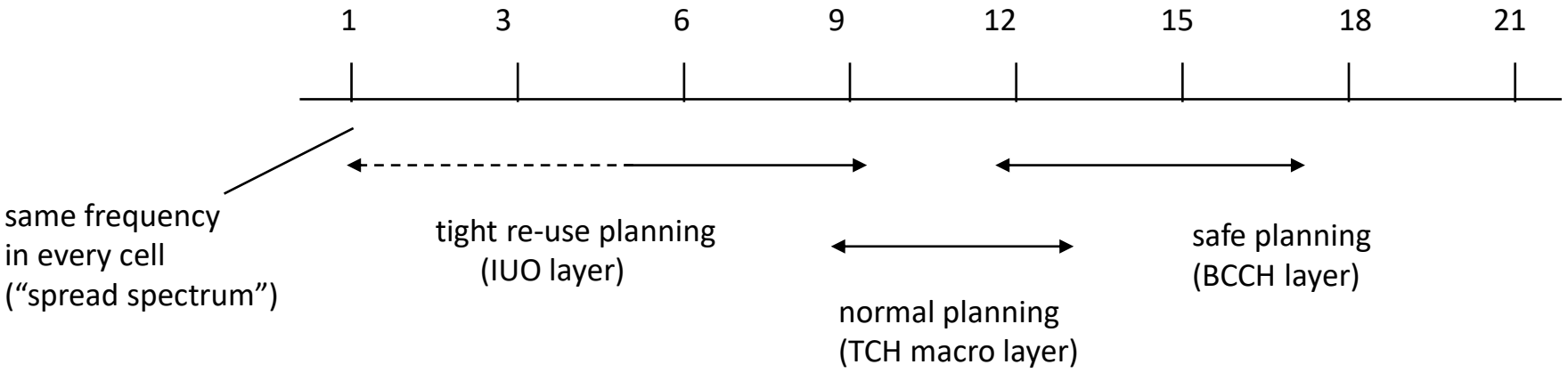
Frequency Plan Best Method

- **Do not use**
  - **Hexagon cell patterns**
  - **Regular grids**
  - **Systematic frequency allocation**
- **Use**
  - **Interference matrix calculation**
  - **Calibrated propagation models**
  - **Minimise total interference in network**



Frequency Plan Re-Use-Factor

- **RuF (Re-use Factor)**
  - **Average number of cells that have different frequencies**
  - **Measure for effectiveness of frequency plan**
  - **Trade-off: effectiveness vs. interferences**
- **Multiple RuFs increase effectiveness of FP**
  - **Compromise between safe, interference free planning and effective resource usage**



Frequency Plan Multiple Re-Use-Factor

- **Capacity increase with multiple RuFs**
  - **e.g. network with 300 cells**
  - **Bandwidth : 8 MHz (40 radio channels)**
- **Single RuF =12**
  - **NW capacity = 40/12 * 300 = 1000 TRX**
- **Multiple RuF**
  - **BCCH layer: re-use =14, (14 frq.)**
  - **Normal TCH: re-use =10, (20 frq.)**
  - **Tight TCH layer: re-use = 6, (6 frq.)**
  - **NW cap. = (1 +2 +1)* 300 = 1200 TRX**

Frequency Plan Constraints

- Co-cell separation
  - e.g. 3 (4 for GSM1800)
  - 600 (800) kHz spacing between frequencies in the same cell
- Co-site separation
  - e.g. 2
  - 400 kHz spacing between frequencies on the same site
- Co-channel interferences from neighbouring sites
- Adjacent channel interferences from neighbouring sites

Frequency Plan Manual Allocation

- With Frequency Groups: 8 groups, 6 ARCFN each

	A1	B1	C1	D1	E1	F1	G1	H1	A2	B2	C2	D2
BCCH	1	26	3	28	5	30	7	32	9	34	11	36
TCH	25	2	27	4	29	6	31	8	33	10	35	12

	E2	F2	G2	H2	A3	B3	C3	D3	E3	F3	G3	H3
BCCH	13	38	15	40	17	42	19	44	21	46	23	48
TCH	37	14	39	16	41	18	43	20	45	22	47	24

- With Separated Bands: 10 groups BCCH, 6 TCH, 3 ARCFN each

BCCH														
A1	B1	C1	D1	E1	F1	G1	H1	I1	L1	A2	B2	C2	D2	E2	F2
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
BCCH														TCH	
G2	H2	I2	L2	A3	B3	C3	D3	E3	F3	G3	H3	I3	L3	M3	N3
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
TCH															
O3	P3	Q3	R3	M4	N4	O4	P4	Q4	R4	M5	N5	O5	P5	Q1	R5
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48

Frequency Plan Manual Allocation

Allocation Criteria

- Take into account both:
  - theoretical dominance area and
  - planner's knowledge of the site
- Starting point:
  - critical site or
  - critical area
- "cluster approach"?
- "dynamic" BCCH allocation
- No more than 60-70 sites!!!

Conclusion

- Method 1 is simpler than method 2
- Method 2 is more accurate

(RuFBCCH > RuFTCH, intracell HO)

	simplicity	BCCH		TCH	
		C/I	C/A	C/I	C/A
groups	x		x	x	x
sub-bands		x			